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PUBLIC WATERMAIN INFRASTRUCTURE PLANS

FOR

ZILBER INDUSTRIAL 3 @ LAKEVIEW CORPORATE PARK

108TH STREET & GREEN BAY ROAD (STH "31")

VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

PLANS PREPARED FOR:

ZILBER PROPERTY GROUP SM

710 N. PLANKINGTON AVE.
MILWAUKEE, WI 53203

LEGEND

	EXISTING	PROPOSED
SANITARY SEWER MANHOLE	S	
STORM SEWER MANHOLE	ST	
STORM SEWER CATCH BASIN (ROUND CASTING)		
FENCE POST		
STORM SEWER CATCH BASIN (RECTANGULAR CASTING)		
PRECAST FLARED END SECTION		
CONCRETE HEADWALL		
VALVE BOX		
FIRE HYDRANT		
CLEANOUT		
SANITARY SEWER		
FORCE MAIN		
STORM SEWER		
DRAIN TILE		
WATER MAIN	W	W
FIRE PROTECTION		FP
ELECTRICAL CABLE	E	E
OVERHEAD WIRES	OHW	OHW
GAS MAIN	G	G
TELEPHONE LINE	T	T
UTILITY CROSSING		X
CAUTION EXISTING UTILITIES NEARBY		CAUTION
GRANULAR TRENCH BACKFILL		
LIGHTING		
ELECTRICAL TRANSFORMER OR PEDESTAL	TF	
POWER POLE		
POWER POLE WITH LIGHT		
GUY WIRE		
STREET SIGN		
CONTOUR	749	749
SPOT ELEVATION	×(750.00)	✦750.00
WETLANDS	WET	
PRIMARY ENVIRONMENTAL CORRIDOR	PEC	
FLOODWAY		
FLOODPLAIN		
HIGH WATER LEVEL (HWL)		
NORMAL WATER LEVEL (NWL)		
DIRECTION OF SURFACE FLOW		
DITCH OR SWALE		
DIVERSION SWALE		
OVERFLOW RELIEF ROUTING		
TREE WITH TRUNK SIZE	6" 6"	
SOIL BORING	B-8 B-9	B-1 B-2
TOPSOIL PROBE		
FENCE LINE, TEMPORARY SILT	SF	SF
FENCE LINE, WIRE		
FENCE LINE, CHAIN LINK OR IRON		
FENCE LINE, WOOD OR PLASTIC		
CONCRETE SIDEWALK		
CURB AND GUTTER		
REVERSE PITCH CURB & GUTTER		
EASEMENT LINE		

ABBREVIATIONS

BL	BASE LINE	NWL	NORMAL WATER LEVEL
C	LONG CHORD OF CURVE	PC	POINT OF CURVATURE
C & G	CURB AND GUTTER	PT	POINT OF TANGENCY
CB	CATCH BASIN	PVI	POINT OF VERTICAL INTERSECTION
CL	CENTERLINE	R	RADIUS
D	DEGREE OF CURVE	ROW	RIGHT-OF-WAY
EP	EDGE OF PAVEMENT	SAN	SANITARY SEWER
FF	FINISHED FLOOR	ST	STORM SEWER
FG	FINISHED GRADE	T	TANGENCY OF CURVE
FL	FLOW LINE	TB	TOP OF BANK
FP	FLOODPLAIN	TC	TOP OF CURB
FR	FRAME	TF	TOP OF FOUNDATION
FW	FLOODWAY	TP	TOP OF PIPE
HWL	HIGH WATER LEVEL	TS	TOP OF SIDEWALK
INV	INVERT	TW	TOP OF WALK
INL	INLET	WM	WATER MAIN
L	LENGTH OF CURVE	△	INTERSECTION ANGLE
MH	MANHOLE		

GOVERNING AGENCY CONTACTS:

PLEASANT PRAIRIE VILLAGE HALL
9915 39TH STREET
PLEASANT PRAIRIE, WI 53158
(262) 694-1400

COMMUNITY DEVELOPMENT DEPARTMENT
ROB HANSON
DIRECTOR OF COMMUNITY DEVELOPMENT
(262) 925-6717
rhanson@pleasantprairiewi.gov

PEGGY HERRICK
ASSISTANT PLANNER & ZONING ADMINISTRATOR
(262) 925-6716
pherrick@pleasantprairiewi.gov

MIRANDA STEINMETZ
ADMINISTRATIVE COORDINATOR / PLANNER I
(262) 925-6717
msteinmetz@pleasantprairiewi.gov

ENGINEERING DEPARTMENT
MATT FINEOUR, P.E.
EXECUTIVE DIRECTOR OF INFRASTRUCTURE MANAGEMENT
(262) 925-6778
mfineour@pleasantprairiewi.gov

KURT DAVIDSEN, P.E.
VILLAGE ENGINEER
(262) 925-6728
kdavidsen@pleasantprairiewi.gov

PUBLIC WORKS DEPARTMENT
TOM HUPP
DIRECTOR OF PUBLIC WORKS
ROGER PRANGE MUNICIPAL BUILDING
8600 GREEN BAY ROAD
(262) 925-6768
thupp@pleasantprairiewi.gov

BUILDING INSPECTION DEPARTMENT
SANDRO PEREZ
BUILDING INSPECTION SUPERINTENDENT
(262) 694-9304 OFFICE
(262) 925-6722 DIRECT
sperez@pleasantprairiewi.gov

MICHAEL KAPRELIAN
BUILDING INSPECTOR
(262) 694-9304
mkaprelian@pleasantprairiewi.gov

FIRE & RESCUE DEPARTMENT
CRAIG ROEPKE,
CHIEF OF FIRE & RESCUE
8044 88TH AVENUE
(262) 694-8027
croepke@pleasantprairiewi.gov

AARON LONGRIE
ASSISTANT FIRE CHIEF
(262) 694-8027
alongrie@pleasantprairiewi.gov

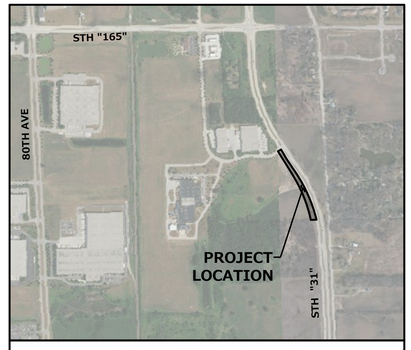
WISCONSIN DOT
ART BAUMANN
TRAFFIC OPERATIONS ENGINEER
(262) 548-6707
art.baumann@dot.wi.gov

KEVIN KOEHNKE, P.E.
PERMITS COORDINATOR
(262) 548-5891
kevin.koehnke@dot.wi.gov

KENOSHA COUNTY
GREG BOLDT, P.E.
DIRECTOR/HIGHWAY COMMISSIONER
DIVISION OF HIGHWAYS - PUBLIC WORKS DEPARTMENT
OFFICE: (262) 857-1870
CELL: (262) 220-0805
gregory.boldt@kenoshacounty.org

WI DEPARTMENT OF NATURAL RESOURCES
ALISON MASEK
WATER MANAGEMENT SPECIALIST
(608) 575-9849
alison.masek@wisconsin.gov

PETER WOOD. P.E.
WATER RESOURCES ENGINEER
(262) 884-2360
peter.wood@wisconsin.gov



LOCATION MAP
SCALE: 1"=1000'

PROJECT TEAM CONTACTS:

CIVIL ENGINEER:
ANDY SHOAF, P.E.
PINNACLE ENGINEERING GROUP
20725 WATERTOWN ROAD, SUITE 100
BROOKFIELD, WI 53186
(262) 754-8888
andrew.shoaf@pinnacle-engr.com

APPLICANT:
JASON LUEDERS
ZILBER PROPERTY GROUP
710 N. PLANKINGTON AVENUE
MILWAUKEE, WI 53203
(414) 274-2807

SURVEYOR:
JOHN KONOPACKI
PINNACLE ENGINEERING GROUP
20725 WATERTOWN ROAD, SUITE 100
BROOKFIELD, WI 53186
(262) 754-8888
john.konopacki@pinnacle-engr.com

ARCHITECT
JEREMY HALL, A.I.A.
PARTNERS IN DESIGN
2610 LAKE COOK ROAD, SUITE 280
RIVERWOODS, IL 60015
(847) 940-0300
jeremyh@pidarchitects.com

PUBLIC UTILITY CONTACTS:

AMERICAN TRANSMISSION COMPANY
BRIAN MCGEE
(262) 506-6895
bmcgee@atcllc.com
(800) 972-5341 EMERGENCY

AT&T
BRENT LEWIS
(414) 429-4214
bl521d@att.com

TDS TELECOM
SOUTHEAST WISCONSIN
(877) 483-7142

TIME WARNER CABLE/SPECTRUM
STEVE CRAMER
CONSTRUCTION SUPERVISOR
(414) 277-4045
steve.cramer@charter.com
(800) 627-2288 EMERGENCY

WEC ENERGY GROUP
JOEL BURROW
CEM MANAGER
(414) 221-3192
joel.burrow@wecenergygroup.com

WE ENERGIES
ALLIE KLAWINSKI
SR. SERVICE MANAGER
(262) 552-3227
allie.klawinski@we-energies.com

MICHAEL BAKER
CUSTOMER SERVICE MANAGER
(262) 886-7041
michael.baker@we-energies.com

HUSSAIN BIYAWERWALA
SENIOR ENGINEER
(262) 886-7074
hussain.biyawerwala@we-energies.com

BRANDON ERTS
GAS ENGINEERING
(414) 221-3654
brandon.erts@we-energies.com

NATURAL GAS EMERGENCY
(800) 261-5325

ELECTRICAL EMERGENCY
(800) 662-4797

VILLAGE NOTES

1. PRIOR TO CONSTRUCTION, A PRE-CONSTRUCTION CONFERENCE MUST BE HELD AT THE VILLAGE OFFICES. THE PRE-CONSTRUCTION CONFERENCE SHALL BE SCHEDULED AND MODERATED BY THE DESIGNING ENGINEER OF RECORD.
2. EACH CONTRACTOR SHALL HAVE A COPY OF THE VILLAGE APPROVED PLANS, PROJECT MANUAL, AND VILLAGE CONSTRUCTION SPECIFICATIONS ON-SITE DURING TIMES OF CONSTRUCTION. THE CONSTRUCTION SPECIFICATIONS ARE AN INTEGRAL PART OF THE CIVIL ENGINEERING PLANS.
3. ALL WORK WITHIN VILLAGE RIGHTS-OF-WAY, PUBLIC EASEMENTS, OR ON VILLAGE FACILITIES SHALL BE COMPLETED IN ACCORDANCE WITH VILLAGE ORDINANCE CHAPTER 405 BY PREQUALIFIED CONTRACTORS AND IN COMPLIANCE WITH THE APPLICABLE WORK IN THE RIGHT OF WAY PERMIT(S). FAILURE TO COMPLETE THE WORK IN ACCORDANCE WITH THESE REQUIREMENTS MAY RESULT IN THE VILLAGE REQUIRING THE ASSOCIATED IMPROVEMENTS TO BE REMOVED AND REPLACED.

INDEX OF SHEETS

C-1	COVER SHEET
C-2	EXISTING CONDITIONS
C-3 - C-4	WATERMAIN PLAN & PROFILE
C-5	EROSION CONTROL PLAN
C-6 - C-8	CONSTRUCTION DETAILS

REQUIRED SUBMITTALS FOR APPROVAL

1. **HOT MIX ASPHALT** - MIX DESIGN
2. **CONCRETE PAVEMENTS (EXTERIOR)** - MIX DESIGN & JOINT PLAN
3. **PAVEMENT STONE BASE COURSE** - GRADATION
4. **PIPE BEDDING & TRENCH BACKFILL** - GRADATION

REQUIRED SUBMITTALS FOR RECORDS

1. **WATER MAIN PIPES, VALVES, HYDRANTS & FITTINGS**
2. **TRACER WIRE**
3. **UNDERGROUND UTILITY LOCATION RECORD DRAWINGS**
4. **CHEMICAL SOIL STABILIZATION MIX DESIGN (IF APPLICABLE)**

GENERAL NOTES

1. THE INTENTION OF THE PLANS AND SPECIFICATIONS IS TO SET FORTH PERFORMANCE AND CONSTRUCTION MATERIAL STANDARDS FOR THE PROPER EXECUTION OF WORK. ALL WORKS CONTAINED WITHIN THE PLANS AND SPECIFICATIONS SHALL BE COMPLETED IN ACCORDANCE WITH ALL REQUIREMENTS FROM LOCAL, STATE, FEDERAL OR OTHER GOVERNING AGENCY'S LAWS, REGULATIONS, JURISDICTIONAL ORDINANCES/CODES/RULES/ETC., AND THE OWNER'S DIRECTION.
2. A GEOTECHNICAL REPORT IS AVAILABLE AT THIS TIME FOR THE PROJECT SITE. IT SHALL BE EXPRESSLY UNDERSTOOD THAT OWNER WILL NOT BE RESPONSIBLE FOR ANY INTERPRETATIONS OR CONCLUSIONS DRAWN THERE FROM BY THE CONTRACTOR. DATA IS MADE AVAILABLE FOR THE CONVENIENCE OF THE CONTRACTOR. THE CONTRACTOR IS RESPONSIBLE FOR PERFORMING ANY ADDITIONAL SOILS INVESTIGATIONS THEY FEEL IS NECESSARY FOR THE PROPER EVALUATION OF THE SITE FOR PURPOSES OF PLANNING, BIDDING, OR CONSTRUCTING THE PROJECT AT NO ADDITIONAL COST TO THE OWNER.
3. THE CONTRACTOR IS RESPONSIBLE TO REVIEW AND UNDERSTAND ALL COMPONENTS OF THE PLANS, REPORTS, AND SPECIFICATIONS, INCLUDING FIELD VERIFYING SOIL CONDITIONS, PRIOR TO SUBMISSION OF A BID PROPOSAL.
4. THE CONTRACTOR SHALL PROMPTLY REPORT ANY ERRORS OR AMBIGUITIES LEARNED AS PART OF THEIR REVIEW OF PLANS, SPECIFICATIONS, REPORTS AND FIELD INVESTIGATIONS.
5. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR THE COMPUTATION OF QUANTITIES AND WORK REQUIRED TO COMPLETE THIS PROJECT. THE CONTRACTOR'S BID SHALL BE BASED ON ITS OWN COMPUTATIONS AND IN NO SUCH INSTANCE RELY ON THE ENGINEER'S ESTIMATE.
6. QUESTIONS/CLARIFICATIONS WILL BE INTERPRETED BY ENGINEER/OWNER PRIOR TO THE AWARD OF CONTRACT. ENGINEER/OWNER WILL SUBMIT OFFICIAL RESPONSES IN WRITING. INTERPRETATIONS PRESENTED IN OFFICIAL RESPONSES SHALL BE BINDING ON ALL PARTIES ASSOCIATED WITH THE CONTRACT. IN NO WAY SHALL WORD-OF-MOUTH DIALOG CONSTITUTE AN OFFICIAL RESPONSE.
7. PRIOR TO CONSTRUCTION, A PRE-CONSTRUCTION CONFERENCE MUST BE HELD AT THE VILLAGE OFFICES. THE PRE-CONSTRUCTION CONFERENCE SHALL BE SCHEDULED AND MODERATED BY THE DESIGNING ENGINEER OF RECORD
8. PRIOR TO START OF WORK, CONTRACTOR SHALL BE COMPLETELY FAMILIAR WITH ALL CONDITIONS OF THE SITE, AND SHALL ACCOUNT FOR CONDITIONS THAT AFFECT, OR MAY AFFECT CONSTRUCTION INCLUDING, BUT NOT LIMITED TO, LIMITATIONS OF WORK ACCESS, SPACE LIMITATIONS, OVERHEAD OBSTRUCTIONS, TRAFFIC PATTERNS, LOCAL REQUIREMENTS, ADJACENT ACTIVITIES, ETC. FAILURE TO CONSIDER SITE CONDITIONS SHALL NOT BE CAUSE FOR CLAIM OF JOB EXTRAS.
9. COMMENCEMENT OF CONSTRUCTION SHALL EXPLICITLY CONFIRM THAT THE CONTRACTOR HAS REVIEWED THE PLANS AND SPECIFICATIONS IN ENTIRETY AND CERTIFIES THAT THEIR SUBMITTED BID PROPOSAL CONTAINS PROVISIONS TO COMPLETE THE PROJECT, WITH THE EXCEPTION OF UNFORESEEN FIELD CONDITIONS; ALL APPLICABLE PERMITS HAVE BEEN OBTAINED; AND CONTRACTOR UNDERSTANDS ALL OF THE REQUIREMENTS OF THE PROJECT.
10. SHOULD ANY DISCREPANCIES OR CONFLICTS IN THE PLANS OR SPECIFICATIONS BE DISCOVERED AFTER THE AWARD OF CONTRACT, ENGINEER SHALL BE NOTIFIED IN WRITING IMMEDIATELY AND CONSTRUCTION OF ITEMS AFFECTED BY THE DISCREPANCIES/CONFLICTS SHALL NOT COMMENCE, OR CONTINUE, UNTIL A WRITTEN RESPONSE FROM ENGINEER/OWNER IS DISTRIBUTED. IN THE EVENT OF A CONFLICT BETWEEN REFERENCED CODES, STANDARDS, SPECIFICATIONS AND PLANS, THE ONE ESTABLISHING THE MOST STRINGENT REQUIREMENTS SHALL BE FOLLOWED.
11. THE CONTRACTOR SHALL, AT ITS OWN EXPENSE, OBTAIN ALL NECESSARY PERMITS AND LICENSES TO COMPLETE THE PROJECT, OBTAINING PERMITS, OR DELAYS, IS NOT CAUSE FOR DELAY OF THE CONTRACT OR SCHEDULE. CONTRACTOR SHALL COMPLY WITH ALL PERMIT REQUIREMENTS.
12. THE CONTRACTOR SHALL NOTIFY ALL INTERESTED GOVERNING AGENCIES. UTILITY COMPANIES AFFECTED BY THIS CONSTRUCTION PROJECT, AND DIGGER'S HOTLINE IN ADVANCE OF CONSTRUCTION TO COMPLY WITH ALL JURISDICTIONAL ORDINANCES/CODES/RULES/ETC., PERMIT STIPULATIONS, AND OTHER APPLICABLE STANDARDS.
13. SAFETY IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR. THE CONTRACTOR SHALL BE RESPONSIBLE TO INITIATE, INSTITUTE, ENFORCE, MAINTAIN, AND SUPERVISE ALL SAFETY PRECAUTIONS AND JOB SITE SAFETY PROGRAMS IN CONNECTION WITH THE WORK.
14. CONTRACTOR SHALL KEEP THE JOBSITE CLEAN AND ORDERLY AT ALL TIMES. ALL LOCATIONS OF THE SITE SHALL BE KEPT IN A WORKING MANNER SUCH THAT DEBRIS IS REMOVED CONTINUOUSLY AND ALL RESPECTIVE CONTRACTORS OPERATE UNDER GENERAL "GOOD HOUSEKEEPING."
15. THE CONTRACTOR SHALL INDEMNIFY THE OWNER, ENGINEER, AND THEIR AGENTS FROM ALL LIABILITY INVOLVED WITH THE CONSTRUCTION, INSTALLATION, AND TESTING OF THE WORK ON THIS PROJECT.
16. ANY WATER SHUTDOWNS NEEDED DURING CONSTRUCTION WILL NEED TO BE PERFORMED BY THE VILLAGE OF PLEASANT PRAIRIE DEPARTMENT OF PUBLIC WORKS AND COORDINATED WITH AREA BUSINESSES.

WISCONSIN
ANDREW J. SHOAF
42919-6
OCONOMOWOC, WI
PROFESSIONAL ENGINEER

11/03/2026

EXPIRATION DATE: JULY 31, 2028

DIGGERS HOTLINE
Toll Free (800) 242-8511
Milwaukee Area (414) 259-1181
Hearing Impaired TDD (800) 542-2289
www.DiggersHotline.com

PINNACLE ENGINEERING GROUP, LLC - ENGINEER'S LIMITATION

PINNACLE ENGINEERING GROUP, LLC AND THEIR CONSULTANTS DO NOT WARRANT OR GUARANTEE THE ACCURACY AND COMPLETENESS OF THE DELIVERABLES HEREIN BEYOND A REASONABLE DILIGENCE. IF ANY MISTAKES, OMISSIONS, OR DISCREPANCIES ARE FOUND TO EXIST WITHIN THE DELIVERABLES, THE ENGINEER SHALL BE PROMPTLY NOTIFIED PRIOR TO BID SO THAT HE MAY HAVE THE OPPORTUNITY TO TAKE WHATEVER STEPS NECESSARY TO RESOLVE THEM. FAILURE TO PROMPTLY NOTIFY THE ENGINEER OF SUCH CONDITIONS SHALL ABSOLVE THE ENGINEER FROM ANY RESPONSIBILITY FOR THE CONSEQUENCES OF SUCH FAILURE. ACTIONS TAKEN WITHOUT THE KNOWLEDGE AND CONSENT TO THE ENGINEER, OR IN CONTRADICTION TO THE ENGINEER'S DELIVERABLES OR RECOMMENDATIONS, SHALL BECOME THE RESPONSIBILITY NOT OF THE ENGINEER BUT OF THE PARTIES RESPONSIBLE FOR TAKING SUCH ACTION.

FURTHERMORE, PINNACLE ENGINEERING GROUP, LLC IS NOT RESPONSIBLE FOR CONSTRUCTION SAFETY OR THE MEANS AND METHODS OF CONSTRUCTION.

REVIEWED:
DESIGNED: AJS
DRAFTED: AJS

PLAN | DESIGN | DELIVER
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PINNACLE ENGINEERING GROUP
ENGINEERING | NATURAL RESOURCES | SURVEYING

WISCONSIN OFFICE:
20725 WATERTOWN ROAD, SUITE 100
BROOKFIELD, WI 53186
(262) 754-8888
CHICAGO | MILWAUKEE · NATIONWIDE

ZILBER INDUSTRIAL 3 @ LAKEVIEW CORPORATE PARK
108TH STREET & GREEN BAY ROAD (STH "31")
VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

COVER SHEET

REVISIONS

1	VOPP SUBMITTAL	05/21/26	
2	VOPP RESUBMITTAL	07/17/26	
3	VOPP RESUBMITTAL	10/02/26	
4	BID SET/VOPP RESUBMITTAL	11/03/26	

PEG JOB No. 644.00-WI
PEG PM AJS
START DATE 03/20/26
SCALE N.T.S.

SHEET C-1 C-8

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COVER SHEET

Z:\PROJECTS\2016\644.00-WI\CAD\SHEETS\WATERMAIN PLAN SHEETS\644.00-WI-SHEETS-COVER.DWG

EXISTING CONDITIONS SURVEY:

EXISTING CONDITIONS SURVEY PROVIDED BY PINNACLE ENGINEERING GROUP. ALTHOUGH PEG HAS NO REASON TO BELIEVE THE SURVEY IS INACCURATE, PEG MAKES NO WARRANTS THAT EXISTING INFORMATION CONTAINED WITHIN THESE PLANS IS ALL-INCLUSIVE OR ACCURATE. CONTRACTOR SHALL UNDERTAKE NECESSARY EFFORTS TO VERIFY THE EXISTING CONDITIONS PRIOR TO THE START OF MATERIAL PROCUREMENT AND CONSTRUCTION EFFORTS/ACTIVITIES.

ALL WORK WITHIN PUBLIC R.O.W. AND PUBLIC EASEMENTS SHALL CONFORM TO THE VILLAGE OF PLEASANT PRAIRIE CHAPTER 405 CONSTRUCTION SPECIFICATIONS.

CONTRACTOR RESPONSIBILITY:

THE INFORMATION SHOWN ON THIS DRAWING CONCERNING TYPE AND LOCATION OF UNDERGROUND UTILITIES IS NOT GUARANTEED TO BE ACCURATE OR ALL INCLUSIVE. THE CONTRACTOR IS RESPONSIBLE FOR MAKING HIS OWN DETERMINATIONS AS TO THE TYPE AND LOCATION OF UNDERGROUND UTILITIES AS MAY BE NECESSARY TO AVOID DAMAGE THERETO. IF ADDITIONAL UTILITIES ARE KNOWN TO EXIST IN THE PROPERTY, THE OWNER WILL PROVIDE EXISTING PLANS OF OTHER UTILITIES SERVING THE SITE AND THE BUILDING THAT OTHERWISE CANNOT BE LOCATED BY A VISUAL OBSERVATION OF THE PROPERTY OR OF WHICH THE SURVEYOR WOULD HAVE NO KNOWLEDGE.

NORTH

GRAPHICAL SCALE (FEET)

0 1" = 60' 120'

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108th STREET
(70' R.O.W.)

N89°47'25"E 267.44'

N78°43'51"E 173.34'

N15°57'24"E 76.07'

S34°42'06"E 180.66'

S59°38'08"E 239.58'

CONCRETE CURB

CONCRETE PAVEMENT

SITE BENCHMARK:
NW FLANGE BOLT
ON HYDRANT
EL.=722.48

SITE BENCHMARK:
NW FLANGE BOLT
ON HYDRANT
EL.=714.89

SANITARY MANHOLE
RIM=717.20
12" W=700.27
12" NE=700.53
12" NE=704.00
8" S=700.38

SANITARY MANHOLE
RIM=714.87
8" N=701.32
8" S=701.27
15" S=708.66

SANITARY MANHOLE
RIM=714.28
12" W & E =699.73
8" S=699.97

SANITARY MANHOLE
RIM=708.83
12" NW=694.51
12" NE=694.56

SANITARY MANHOLE
RIM=722.51
12" SW &
SE=705.31

STORM MANHOLE
RIM=722.05
15" NW & SE=718.90

STORM MANHOLE
RIM=722.36
15" NW & SE=718.88

STORM MANHOLE
RIM=713.16
18" W=708.21
15" N=708.46
15" S=708.26
18" E=708.21

STORM MANHOLE
RIM=716.01
18" W=710.76
15" N=710.98
15" S=711.08

INLET
RIM=715.76
15" S=711.51

INLET
RIM=715.16
15" N=711.14

INLET
RIM=708.30
15" S=704.09

INLET
RIM=708.28
15" S & N=703.91

INLET
RIM=712.69
15" N=708.19

INLET
RIM=724.30

INLET
RIM=721.30

INLET
RIM=721.66

INLET
RIM=720.74
42" INV.=713.97

INLET
RIM=720.13

INLET
RIM=720.14
42" INV.=713.80

INLET
RIM=720.51

INLET
RIM=722.80

INLET
RIM=723.81

INLET
RIM=729.63
36" W&E=719.92

INLET
RIM=729.20

INLET
RIM=730.13

15" RCP
INV.=719.34

RAMP

15" RCP
INV.=718.91

42" RCP
INV.=714.45

42" RCP
INV.=713.61

12" RCP
INV.=718.19

36" RCP
INV.=719.20

36" RCP
INV.=721.13

6" PVC
INV.=723.07

12" RCP
INV.=724.97

4" PVC
W INV.=709.57
E INV.=710.41

18" CMP
INV.=706.67

18" CMP
INV.=704.70

9" CLAY CULVERT
SW INV.=706.20
NE INV.=706.52

VERTICAL 8" PVC
PIPE WITH CAGE
RIM=712.39
INV.=709.3

FOUND 1/2"
IRON REBAR

FOUND 3/4"
IRON PIPE

FOUND 1" IRON PIPE WITH CAP
AND CAPPED 1/2" IRON REBAR
ON SECTION LINE, 0.32' EAST

20' GENERAL UTILITY EASEMENT
PER C.S.M. NO. 2273

NO ACCESS PER
C.S.M. NO. 2273

50' x 500' VISION CORNER
EASEMENT PER C.S.M. NO. 2273

100 YEAR FLOOD
PER FEMA MAP
NO. 55059C0194D
DATED JUNE 19, 2012

WETLANDS DELINEATED BY
LANDCO2 APRIL 2026

WETLAND DATA
POINT (TYP.)

EDGE OF
WATER

TREE LINE

FIELD
CROSSING

CENTERLINE
DITCH

EDGE OF
GRAVEL

MARKED
FORCED
MAIN

CONCRETE CURB

CONCRETE CURB

TP-1

TP-2

TP-3

TP-4

TP-5

TP-6

TP-7

TP-8

TP-9

PARCEL 9
995,257 SQ. FT.
22.8480 ACRES

- VACANT -
NO BUILDINGS

PARCEL 2
C.S.M. NO. 1971
TAX KEY #92-4-122-273-0204

N89°28'52"E 2658.73'
SOUTH LINE OF THE NW 1/4 SEC. 27, T1N, R22E

S89°28'52"W 1134.14'

9°55'24"W 330.00'

1524.59' SE CORNER, NW 1/4 SEC. 27, T1N, R22E
(CONC. MON. W/ BRASS CAP)

REFERENCE BENCHMARK:
CONC. MON. W/ BRASS CAP
EL.=704.98

SW CORNER, NW 1/4 SEC. 27, T1N, R22E
SE CORNER, NE 1/4 SEC. 28, T1N, R22E
(CONC. MON. W/ BRASS CAP)
P.O.C. & P.O.B.

CENTERLINE
DITCH

EAST LINE OF
SEC. 28

WEST LINE
SEC. 27
N03°22'44"E

REVIEWED:

DESIGNED: AJS

DRAFTED: AJS

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WISCONSIN OFFICE:
20725 WATERTOWN ROAD, SUITE 100
BROOKFIELD, WI 53186
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VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

EXISTING CONDITIONS

REVISIONS

1	VOPP SUBMITTAL	05/21/26
2	VOPP RESUBMITTAL	07/17/26
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4	BID SET/VOPP RESUBMITTAL	11/03/26

PEG JOB No. 644.00-WI
AJS
PES PM
START DATE 03/20/26
SCALE

SHEET
C-2
C-8

EXISTING CONDITIONS

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CONTRACTOR SHALL VERIFY LOCATION AND ELEVATION OF ALL EXISTING UTILITIES PRIOR TO INSTALLATION OF PROPOSED UTILITIES.

ALL WORK WITHIN PUBLIC R.O.W. AND PUBLIC EASEMENTS SHALL CONFORM TO THE VILLAGE OF PLEASANT PRAIRIE CHAPTER 405 CONSTRUCTION SPECIFICATIONS.

PUBLIC WORKS COORDINATION

1. CONTRACTOR SHALL COORDINATE AND SCHEDULE INSPECTIONS FOR ALL WORK ON PUBLIC IMPROVEMENTS WITH THE ENGINEERING DEPARTMENT. WORK WITHIN PUBLIC EASEMENTS OR RIGHT OF WAY SHALL NOT COMMENCE UNTIL ENGINEERING DEPARTMENT HAS BEEN CONTACTED AND INSPECTOR IS ON SITE.
2. ALL PUBLIC WATER VALVES SHALL BE OPERATED BY OR UNDER THE SUPERVISION OF THE PUBLIC WORKS DEPARTMENT. CONTRACTOR SHALL COORDINATE WITH THE PUBLIC WORKS DEPARTMENT.

EXISTING CONDITIONS SURVEY:

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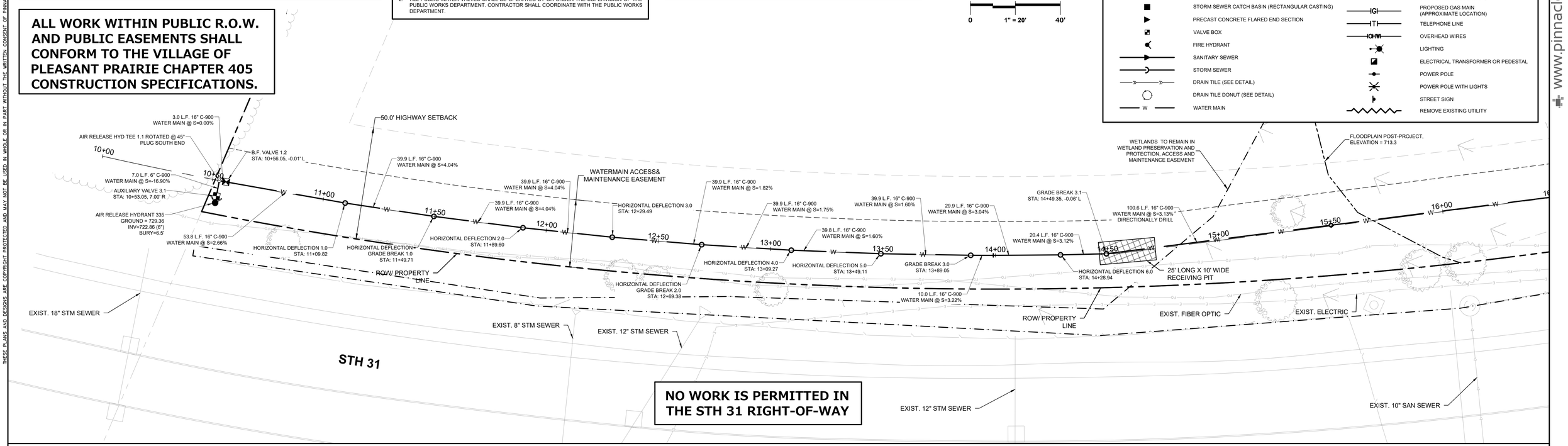
NORTH

GRAPHICAL SCALE (FEET)

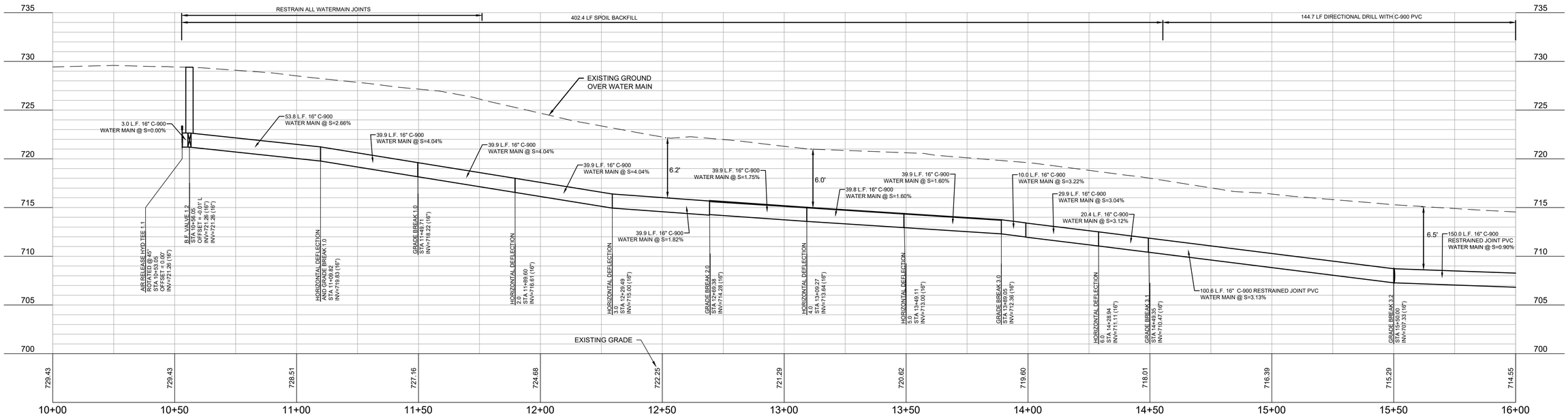
0 1" = 20' 40'

LEGEND

- SANITARY SEWER MANHOLE
- STORM SEWER MANHOLE
- STORM SEWER CATCH BASIN (ROUND CASTING)
- STORM SEWER CATCH BASIN (RECTANGULAR CASTING)
- PRECAST CONCRETE FLARED END SECTION
- VALVE BOX
- FIRE HYDRANT
- SANITARY SEWER
- STORM SEWER
- DRAIN TILE (SEE DETAIL)
- DRAIN TILE DONUT (SEE DETAIL)
- W WATER MAIN
- FP FIRE PROTECTION
- UTILITY CROSSING
- PROPOSED ELECTRICAL CABLE (APPROXIMATE LOCATION)
- PROPOSED GAS MAIN (APPROXIMATE LOCATION)
- TELEPHONE LINE
- OHW OVERHEAD WIRES
- LIGHTING
- ELECTRICAL TRANSFORMER OR PEDESTAL
- POWER POLE
- POWER POLE WITH LIGHTS
- STREET SIGN
- REMOVE EXISTING UTILITY



NO WORK IS PERMITTED IN THE STH 31 RIGHT-OF-WAY



REVIEWED:
DESIGNED: AJS
DRAFTED: AJS

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WISCONSIN OFFICE:
20725 WATERTOWN ROAD, SUITE 100
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VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

WATERMAIN PLAN & PROFILE

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WATERMAIN PLAN & PROFILE

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- FIRE PROTECTION
- UTILITY CROSSING
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- PROPOSED GAS MAIN (APPROXIMATE LOCATION)
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- OVERHEAD WIRES
- LIGHTING
- ELECTRICAL TRANSFORMER OR PEDESTAL
- POWER POLE
- POWER POLE WITH LIGHTS
- STREET SIGN
- REMOVE EXISTING UTILITY

NORTH

GRAPHICAL SCALE (FEET)

0 1" = 20' 40'

FLOODPLAIN POST-PROJECT, ELEVATION = 713.3

WETLANDS TO REMAIN IN WETLAND PRESERVATION AND PROTECTION, ACCESS AND MAINTENANCE EASEMENT

50' x 500' VISION CORNER EASEMENT PER C.S.M. NO. 2273

25' LONG BY 10' WIDE BORE PIT

WATERMAIN ACCESS & MAINTENANCE EASEMENT

189.4 L.F. 16" C-900 WATER MAIN @ S=-1.98%

11.1 L.F. 16" C-900 WATER MAIN @ S=-1.98%

133.7 L.F. 16" C-900 WATER MAIN @ S=-0.21%

5.0 L.F. 16" C-900 WATER MAIN @ S=0.00%

143.0 L.F. 10" PVC @ S=0.92%

GRADE BREAK 3.3 STA: 17+00.00

HORIZONTAL DEFLECTION 8.0 STA: 17+96.96

GRADE BREAK 8.0 STA: 19+97.48

HORIZONTAL DEFLECTION 9.0 STA: 19+86.36

20+91.14 0.0

B.F. VALVE 2.2 STA: 21+31.13, 0.00'

B.F. VALVE 10.0 STA: 21+36.13, 5.00' R INV = 712.17 (16")

CONNECT TO EXIST INV=712.17 (16" SE) INV=712.17 (16" NE)

SAN MH (14.0) 48" DIA RIM=717.22 INV=704.00 (10" NE)

SAN MH (15.0) 48" DIA RIM=722.50 INV=705.31 (10" SW)

108TH ST

PROPERTY LINE

SAWCUT

CAUTION

EXIST. FIBER OPTIC

EXIST. 10" SAN SEWER

EXIST. 10" SAN SEWER

EXIST. 36" STM SEWER

ROW/ PROPERTY LINE

EXIST. FIBER OPTIC

EXIST. ELECTRIC

EXIST. 10" SAN SEWER

WETLANDS TO REMAIN IN WETLAND PRESERVATION AND PROTECTION, ACCESS AND MAINTENANCE EASEMENT

EXIST. 16" WATERMAIN

EXIST. 15" STM SEWER

REMOVE AND REPLACE 30" CONCRETE CURB & GUTTER TO NEAREST JOINT

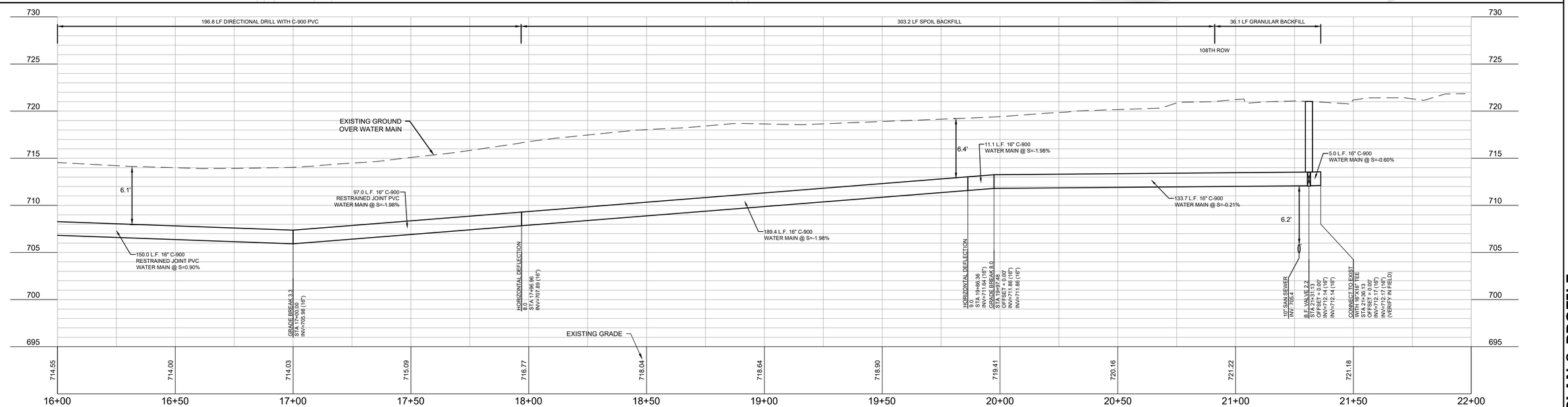
REMOVE AND REPLACE CONCRETE PAVEMENT. REQUIRE HIGH EARLY STRENGTH CONCRETE FOR PATCH. PATCH IN SECTIONS TO KEEP ROADWAY INTERSECTION OPEN.

105 LF SAWCUT EXISTING PAVEMENT

STH 31

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DESIGNED: AJS
DRAFTED: AJS

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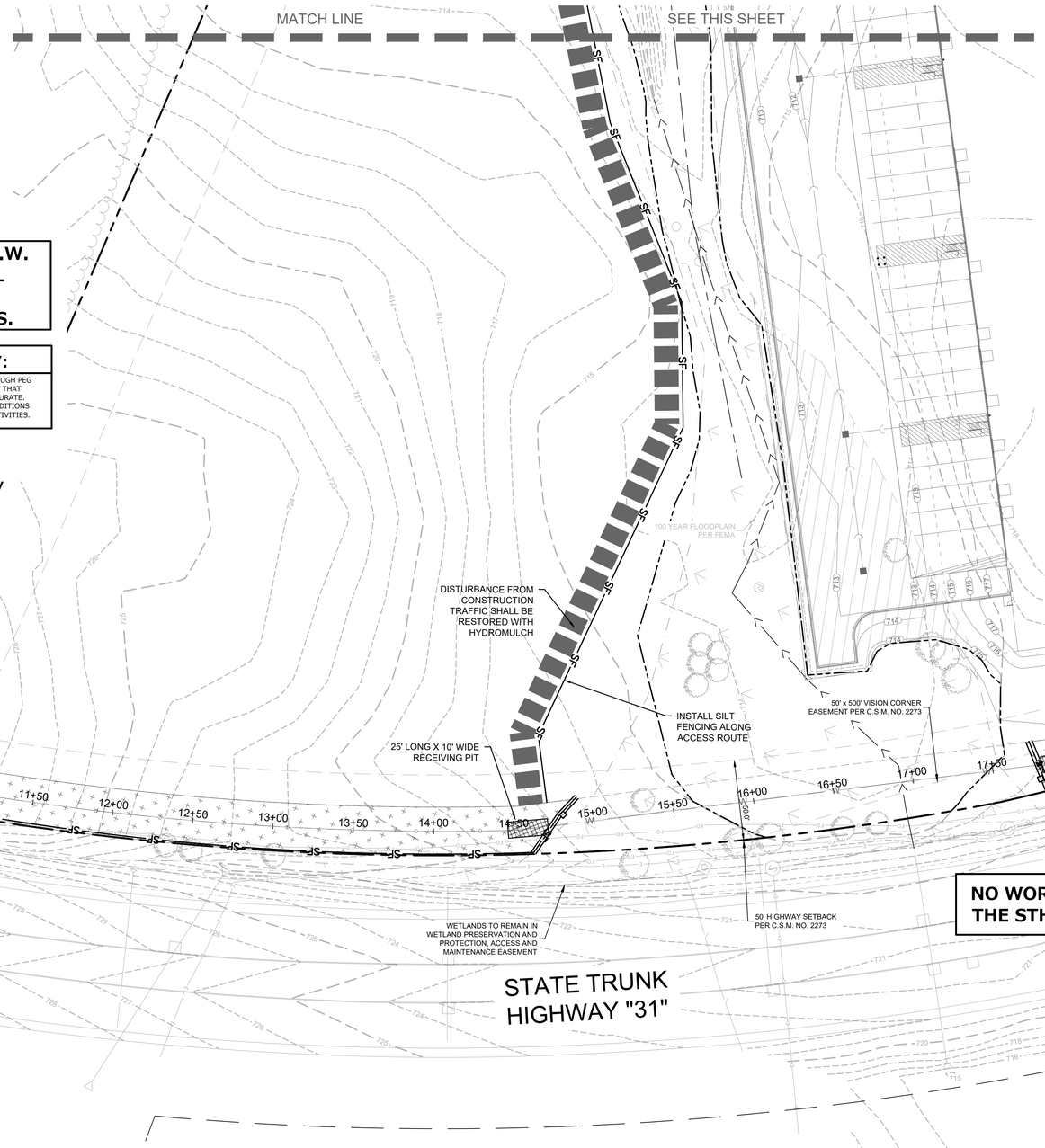
LEGEND

- STORM SEWER MANHOLE
- STORM STORM CATCH BASIN (ROUND CASTING)
- STORM SEWER CATCH BASIN (RECTANGULAR CASTING)
- PRECAST FLARED END SECTION
- PROPOSED CONTOUR
- WETLANDS
- FLOODPLAIN
- NORMAL WATER LEVEL (NWL)
- SF — SILT FENCE
- ORANGE SAFETY FENCE
- INLET PROTECTION (TYPE B)
- INLET PROTECTION (TYPE C)
- MEDIUM RIP RAP
- STRAW WADDLE
- CULVERT CHECK
- HYDROSEED (PER MANUFACTURER SPECIFICATIONS)
- EROSION CONTROL BLANKET (NORTH AMERICAN GREEN S75 OR EQUAL)
- TURF REINFORCEMENT MAT (NORTH AMERICAN GREEN TMAX OR EQUAL)

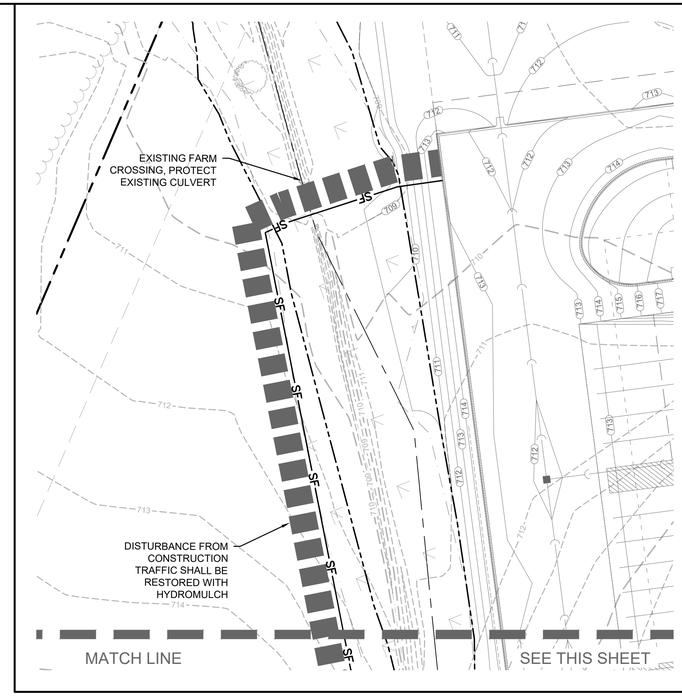
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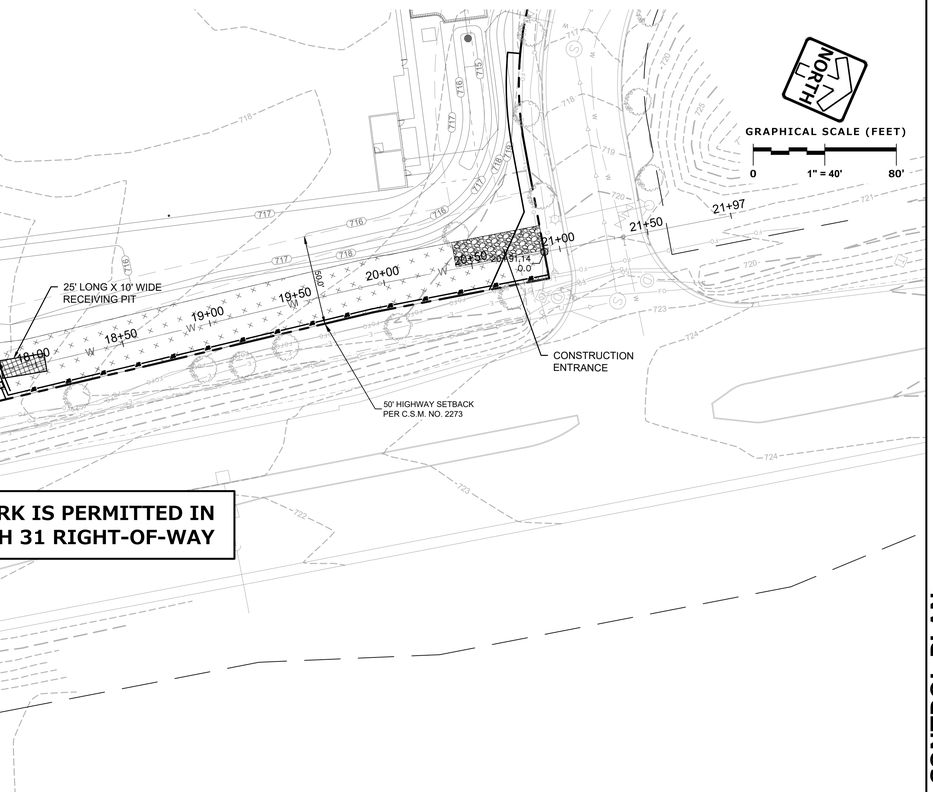
NO WORK IS PERMITTED IN THE STH 31 RIGHT-OF-WAY



NORTH

GRAPHICAL SCALE (FEET)

0 1" = 40' 80'



NORTH

GRAPHICAL SCALE (FEET)

0 1" = 40' 80'

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EROSION CONTROL PLAN

REVIEWED:

DESIGNED: AJS

DRAFTED: AJS

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EROSION CONTROL PLAN

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1.0 POTENTIAL POLLUTANT SOURCES

GIVEN THE PROPOSED ACTIVITY ON THE PROJECT SITE, THE PRIMARY POTENTIAL POLLUTANT SOURCE ASSOCIATED WITH THIS CONSTRUCTION PROJECT IS SOIL EROSION AND TRANSPORTATION; REFER TO SECTION 4 OF THIS PLAN. ADDITIONAL POTENTIAL SOURCES OF POLLUTION MAY INCLUDE: FUEL TANKS, WASTE CONTAINERS, OIL OR OTHER PETROLEUM PRODUCTS, DETERGENTS, PAINTS, CONSTRUCTION DEBRIS, SANITARY STATIONS, FERTILIZERS, AND DUST; REFER TO SECTION 5 OF THIS PLAN.

2.0 EROSION AND SEDIMENT CONTROL IMPLEMENTATION

THE FOLLOWING ARE DESCRIPTIONS OF THE EROSION AND SEDIMENT CONTROL PRACTICES THAT SHALL BE IMPLEMENTED DURING CONSTRUCTION OF THIS PROJECT. IN ADDITION TO THESE MEASURES, CONTRACTOR SHALL DISTURB ONLY AREAS NECESSARY TO COMPLETE THE CONSTRUCTION PROJECT. ALL PRACTICES SHALL BE CONDUCTED IN ACCORDANCE WITH THE BEST MANAGEMENT PRACTICES (BMP).

2.1 CONSTRUCTION AND EROSION CONTROL SEQUENCING

CONSTRUCTION SEQUENCING WILL BE UTILIZED AS A MEANS OF CONTROLLING EROSION AND LIMITING SEDIMENT TRANSPORT. SEQUENCING AS LISTED BELOW IS GENERAL IN NATURE AND MAY VARY DEPENDING ON WEATHER CONDITIONS AND/OR PHASING OF CONSTRUCTION. THE CONTRACTOR SHALL SUBMIT A DETAILED SITE SEQUENCING PLAN TO OWNER FOR APPROVAL AT LEAST 5 BUSINESS DAYS PRIOR TO COMMENCEMENT OF ANY CONSTRUCTION ACTIVITIES. CONTRACTOR MAY MODIFY SEQUENCING AFTER ITEM 6 AS NEEDED TO COMPLETE CONSTRUCTION ONLY IF EROSION CONTROLS ARE MAINTAINED IN ACCORDANCE WITH THE CONSTRUCTION SITE EROSION AND SEDIMENT CONTROL REQUIREMENTS.

1. INSTALL TEMPORARY CONSTRUCTION ENTRANCES, INLET PROTECTION ON EXISTING STORM SEWER AND CULVERT INLET LOCATIONS, AND PERIMETER SILT FENCING.
2. INSTALL SILT FENCING ALONG THE PERIMETER OF PROPOSED TOPSOIL STOCKPILE LOCATIONS. THE FIRST TOPSOIL DEPOSITED WITHIN THE STOCKPILE LIMITS SHALL BE PLACED TO CREATE TEMPORARY BERMING ALONG THE SILT FENCE TO PREVENT DIRECT STORMWATER RUNOFF AGAINST SILT FENCING. CONTRACTOR SHALL LIMIT LAND DISTURBING ACTIVITIES ASSOCIATED WITH TEMPORARY BERMING TO A MINIMUM.
3. STRIP TOPSOIL WITHIN THE LIMITS OF THE SEDIMENT TRAPS THAT WILL BE USED FOR TEMPORARY SEDIMENT CONTROL. STRIPPED TOPSOIL SHALL BE PLACED TO CONSTRUCT DIVERSION BERMING OR PLACED WITHIN THE STOCKPILE LIMITS.
4. STRIP TOPSOIL ALONG THE REMAINDER OF DIVERSION BERMING AND IMMEDIATELY PLACE TOPSOIL TO CREATE THE BERMING. MASS TOPSOIL STRIPPING SHALL NOT OCCUR UNTIL ALL DOWNSTREAM SEDIMENT CONTROLS ARE IN PLACE.
5. CONDUCT ROUGH GRADING OPERATIONS AND UTILITY PIPING INSTALLATION. DRAIN TILE SHALL NOT BE INSTALLED UNTIL UPLAND AREAS CONTRIBUTING STORMWATER RUNOFF ARE STABILIZED. DITCH CHECKS SHALL BE INSTALLED WITHIN DRAINAGE DITCHES IMMEDIATELY FOLLOWING CREATION OF DITCHES AND INLET PROTECTION SHALL BE INSTALLED TO PROTECT ANY STORM SEWER OR CULVERTS THAT WILL FUNCTION DURING CONSTRUCTION.
6. FINE GRADE SUB-GRADE SOILS WITHIN PAVEMENT AND BUILDING LIMITS. PLACE STONE BASE MATERIAL AS SOON AS POSSIBLE FOLLOWING COMPLETION OF FINE GRADING EFFORTS.
7. FINE GRADE REMAINING DISTURBED AREAS. PLACE SALVAGED TOPSOIL, EROSION BLANKETS/MATTING, AND SEED/MULCH AS SOON AS POSSIBLE FOLLOWING COMPLETION OF FINE GRADING EFFORTS.
8. EROSION CONTROLS SHALL NOT BE REMOVED UNTIL SITE IS FULLY STABILIZED OR 70% VEGETATIVE COVER IS ESTABLISHED. CONTRACTOR IS RESPONSIBLE FOR REMOVAL OF SILT FENCE, TEMPORARY FENCING/PRETECTION, DITCH CHECKS, AND OTHER TEMPORARY CONTROLS, AND RESTORATION PRACTICES AS NECESSARY, TO THE SATISFACTION OF THE OWNER.

2.2 STABILIZATION PRACTICES

THE DATES WHEN MAJOR GRADING ACTIVITIES OCCUR, WHEN CONSTRUCTION ACTIVITIES TEMPORARILY OR PERMANENTLY CEASE ON A PORTION OF THE SITE, AND WHEN STABILIZATION MEASURES ARE INITIATED, SHALL BE RECORDED ON THE STABILIZATION SCHEDULE FOR MAJOR GRADING ACTIVITIES.

STABILIZATION MEASURES SHALL BE INITIATED AS SOON AS PRACTICABLE IN PORTIONS OF THE SITE WHERE CONSTRUCTION ACTIVITIES HAVE TEMPORARILY OR PERMANENTLY CEASED. NO MORE THAN SEVEN (7) DAYS SHALL PASS AFTER THE CONSTRUCTION ACTIVITY IN THAT PORTION OF THE SITE HAS PERMANENTLY CEASED UNLESS:

THE INITIATION OF STABILIZATION MEASURES BY THE SEVENTH (7) DAY AFTER CONSTRUCTION ACTIVITY TEMPORARILY OR PERMANENTLY CEASE IS PRECLUDED BY SNOW COVER. IN THAT EVENT, STABILIZATION MEASURE SHALL BE INITIATED AS SOON AS PRACTICABLE.

CONSTRUCTION ACTIVITY WILL RESUME ON A PORTION OF THE SITE WITHIN FOURTEEN (14) DAYS FROM WHEN ACTIVITIES CEASED, (I.E. THE TOTAL TIME PERIOD THAT THE CONSTRUCTION ACTIVITY IS TEMPORARILY CEASED IS LESS THAN FOURTEEN (14) DAYS). IN THAT EVENT, STABILIZATION MEASURES DO NOT HAVE TO BE INITIATED ON THAT PORTION OF THE SITE BY THE SEVENTH (7) DAY AFTER CONSTRUCTION ACTIVITY HAS TEMPORARILY CEASED. SEE THE SOIL PROTECTION CHART PRESENTED IN THE CONSTRUCTION DOCUMENTS FOR RATES OF PERMANENT AND TEMPORARY VEGETATION.

STABILIZATION MEASURES SHALL BE DETERMINED BASED ON SITE CONDITIONS AT THE TIME CONSTRUCTION ACTIVITY HAS CEASED, INCLUDING BUT NOT LIMITED TO WEATHER CONDITIONS AND LENGTH OF TIME MEASURE MUST BE EFFECTIVE. THE FOLLOWING ARE ACCEPTABLE STABILIZATION MEASURES.

PERMANENT SEEDING; IN ACCORDANCE WITH APPROVED LANDSCAPING PLAN
TEMPORARY SEEDING MAY CONSIST OF SPRING OATS (100LBS/ACRE) AND/OR WHEAT OR CEREAL RYE (150LBS/ACRE)
HYDRO-MULCHING WITH A TACKIFIER
GEOTEXTILE EROSION MATTING
SODDING

CONTROL MEASURE GROUP	CONTROL MEASURE	CONTROL MEASURE CHARACTERISTICS
VEGETATIVE SOIL COVER	TEMPORARY SEEDING	PROVIDES QUICK TEMPORARY COVER TO CONTROL EROSION WHEN PERMANENT SEEDING IS NOT DESIRED OR TIME OF YEAR IS INAPPROPRIATE.
	PERMANENT SEEDING	PROVIDES PERMANENT VEGETATIVE COVER TO CONTROL EROSION, FILTERS SEDIMENT FROM WATER, MAY BE PART OF FINAL LANDSCAPE PLAN.
NON VEGETATIVE SOIL COVER	AGGREGATE COVER	PROVIDES TEMPORARY COVER ON ROADS AND PARKING LOTS AND AREAS WHERE VEGETATION CANNOT BE ESTABLISHED, PREVENTS MUD FROM BEING PICKED UP AND TRANSPORTED OFF-SITE.
	PAVING	PROVIDES PERMANENT COVER ON PARKING LOTS AND ROADS OR OTHER AREAS WHERE VEGETATION CANNOT BE ESTABLISHED.
DIVERSIONS	DIVERSION BERM / SWALE	DIVERTS RUNOFF TO A SEDIMENT TRAP OR OTHER CONTROL.
ENCLOSED DRAINAGE	STORM SEWER	CONVEYS SEDIMENT LADEN WATER TO A SEDIMENT BASIN.
OUTLETS	APRON ENDWALL OR RIPRAP	PROTECTS DOWNSTREAM CHANNEL FROM HIGH VELOCITY OF FLOW DISCHARGING FROM STRUCTURE.
SEDIMENT BASINS	TEMPORARY SEDIMENT TRAP	CONSTRUCTED TO REMOVE SILTATION FROM RUNOFF FROM SITE DIVERSION BERMS/SWALES AND IN OVERLAND FLOOD ROUTE. CAN BE CONVERTED TO PERMANENT SEDIMENT BASIN.
SEDIMENT FILTERS	SILT FENCE	PLACED DOWN SLOPE OF DISTURBED AREA TO KEEP RUNOFF CONTAINED ON-SITE.
	INLET PROTECTION	INSTALLED IN OPEN GRATE STRUCTURES TO COLLECT SEDIMENT.
	DITCH CHECK	PLACED IN DRAINAGE CHANNELS TO FILTER SEDIMENT FROM RUNOFF.
MUD AND DUST CONTROL	CONSTRUCTION ENTRANCE	REDUCES SOIL EROSION POLLUTANTS BEING TRANSPORTED OFF-SITE.
	STREET SWEEPING	REDUCES POLLUTANTS TRACKED FROM CONSTRUCTION SITE.
	DUST CONTROL	PREVENTS DUST FROM LEAVING CONSTRUCTION SITE.

STABILIZATION EFFECTIVENESS (TIME OF YEAR)

STABILIZATION TYPE	STABILIZATION UTILIZATION PERIODS											
	JAN.	FEB.	MAR.	APR.	MAY	JUNE	JULY	AUG.	SEPT.	OCT.	NOV.	DEC.
PERMANENT SEEDING			A	*	*	*	*	*	*			
DORMANT SEEDING	B										B	
TEMPORARY SEEDING			C	*	*	*	D *	*	*			
SODDING			E	*	*	*	*	*	*			

A, KENTUCKY BLUEGRASS 90 LBS/ACRE MIXED WITH PERENNIAL RYEGRASS 30 LBS/ACRE.
B, KENTUCKY BLUEGRASS 135 LBS/ACRE MIXED WITH PERENNIAL RYEGRASS 45 LBS/ACRE + 2 TONS STRAW MULCH/ACRE.
C, SPRING OATS 100 LBS/ACRE.
D, WHEAT OR CEREAL RYE 150 LBS/ACRE.
E, SOD.
F, STRAW MULCH 2 TONS/ACRE.
* IRRIGATION/WATERING REQUIRED TO SUPPORT ESTABLISHMENT AS NEEDED.

2.3 STRUCTURAL PRACTICES

THE FOLLOWING ARE DESCRIPTIONS OF STRUCTURAL PRACTICES TO BE IMPLEMENTED TO DIVERT FLOWS FROM EXPOSED SOILS, STORE FLOWS, OR OTHERWISE LIMIT THE DISCHARGE OF POLLUTANTS FROM EXPOSED AREAS OF THE SITE INCLUDING THE PROPOSED AND EXISTING WETLAND AREAS. SUCH PRACTICES COULD INCLUDE SILT FENCE, PROTECTION FENCE, CONSTRUCTION ENTRANCE, DITCH CHECK, EROSION CONTROL MATTING, DIVERSION BERM/SWALE, SEDIMENT TRAP, LEVEL SPREADER, INLET PROTECTION, OUTLET PROTECTION, AND TEMPORARY OR PERMANENT SEDIMENT BASIN. THE FOLLOWING STRUCTURAL PRACTICES ARE TO BE UTILIZED DURING THIS PROJECT.

SILT FENCE SHALL BE PLACED DOWN SLOPE OF DISTURBED AREAS OF THE CONSTRUCTION SITE AND AROUND THE PERIMETER OF THE TOPSOIL STOCKPILE. THIS INCLUDES PROTECTION OF EXISTING WETLAND AREAS TO BE MAINTAINED. SILT FENCE MAY ALSO BE USED AS A TEMPORARY CONTROL DEVICE WHERE SEDIMENTATION RUNOFF IS DISCOVERED.

CONSTRUCTION ENTRANCE SHALL BE INSTALLED TO REDUCE SOIL EROSION POLLUTANTS FROM LEAVING THE SITE DURING CONSTRUCTION ACTIVITIES. IF THE CRUSHED STONE DOES NOT ADEQUATELY REMOVE MUD FROM VEHICLE TIRES, THEY SHALL BE HOSED OFF BEFORE ENTERING A PAVED ROADWAY. ANY SOIL DEPOSITED ON THE PUBLIC PAVED ROAD WAY SHALL BE REMOVED IMMEDIATELY.

DITCH CHECK (STRAW BALES) SHALL BE INSTALLED IN DRAINAGE CHANNELS AS NEEDED.

EROSION CONTROL MATTING SHALL BE PLACED ON AREAS OR EMBANKMENTS HAVING SLOPES GREATER THAN OR EQUAL TO 3H:1V, BEFORE VEGETATION IS ESTABLISHED.

DIVERSION BERM/SWALE SHALL BE CONSTRUCTED TO DIVERT RUNOFF AROUND THE SITE AND TO DIVERT RUNOFF FROM THE DISTURBED AREA TO A SEDIMENT TRAP OR OTHER CONTROL. BERMS/SWALES SHALL BE STABILIZED WITH EQUIPMENT TRACKING AND TEMPORARY SEEDING.

SEDIMENT TRAPS/BASIN SHALL BE CONSTRUCTED TO COLLECT RUNOFF AND RUNOFF FROM SITE DIVERSION BERMS/SWALES.

INLET PROTECTION SHALL BE INSTALLED AT STORMWATER DRAINAGE INLETS TO REDUCE SEDIMENT WITHIN STORM SEWER CONVEYANCE FEATURES.

OUTLET SCOUR PROTECTION SHALL BE INSTALLED AT STORMWATER DRAINAGE OUTLETS TO DIFFUSE FLOWS.

3.0 ADDITIONAL PRACTICES

ADDITIONAL POLLUTANT CONTROL MEASURES TO BE IMPLEMENTED DURING CONSTRUCTION ACTIVITIES SHALL INCLUDE, BUT NOT BE LIMITED TO THE FOLLOWING.

CONSTRUCTION WASTE SHALL BE PROPERLY DISPOSED OF. THIS INCLUDES ALL CONSTRUCTION SITE WASTE MATERIAL, SANITARY WASTE, AND WASTE FROM VEHICLE TRACKING OF SEDIMENTS. THE CONTRACTOR SHALL ENSURE THAT NO MATERIAL WASTES OR UNUSED BUILDING MATERIALS SHALL BE BURIED, DUMPED, BURNED, OR DISCHARGED TO THE WATERS OF THE STATE. VEHICLES HAULING MATERIAL AWAY FROM THE SITE SHALL BE COVERED WITH A TARPAULIN TO PREVENT BLOWING DEBRIS.

DUST CONTROL SHALL BE ACCOMPLISHED BY ONE OR MORE OF THE FOLLOWING METHODS: COVERING 30% OR MORE OF THE SOIL SURFACE WITH A NON-ERODIBLE MATERIAL.

ROUGHENING (EQUIPMENT TRACKING) THE SOIL TO PRODUCE RIDGES PERPENDICULAR TO THE PREVAILING WIND. RIDGES SHALL BE AT LEAST SIX (6) INCHES IN HEIGHT.

FREQUENT WATERING OF EXCAVATION AND FILL AREAS.

PROVIDING GRAVEL OR PAVING AT ENTRANCE/EXIT DRIVES, PARKING AREAS AND TRANSIT PATHS.

STREET SWEEPING SHALL BE PERFORMED TO IMMEDIATELY REMOVE ANY SEDIMENT TRACKED ON PAVEMENTS.

4.0 EROSION AND SEDIMENT STRUCTURAL PRACTICE MAINTENANCE

THE FOLLOWING MAINTENANCE PRACTICES SHALL BE USED TO MAINTAIN, IN GOOD AND EFFECTIVE OPERATING CONDITIONS, VEGETATION, EROSION AND SEDIMENT CONTROL MEASURES, AND OTHER PROTECTIVE MEASURES IDENTIFIED IN THIS PLAN. UPON IDENTIFICATION, DEFICIENCIES IN STORMWATER CONTROLS SHALL BE ADDRESSED IMMEDIATELY. THE MAINTENANCE PROCEDURES FOR THIS DEVELOPMENT SHALL INCLUDE, BUT NOT BE LIMITED TO THE BELOW.

SILT FENCE - REPAIR OR REPLACE ANY DAMAGED FILTER FABRIC AND/OR STAKES. REMOVE ACCUMULATED SEDIMENT WHEN IT HAS REACHED ONE-HALF THE ABOVE GROUND HEIGHT OF THE FENCE.

CONSTRUCTION ENTRANCE - AS NEEDED, ADD STONE TO MAINTAIN CONSTRUCTION ENTRANCE DIMENSIONS AND EFFECTIVENESS.

DITCH CHECK (STRAW BALES) - RE-SECURE STAKES; ADJUST OR REPOSITION BALES TO ADDRESS PROPER FLOW OF STORMWATER; AND REMOVE ACCUMULATED SEDIMENT WHEN IT HAS REACHED ONE-HALF THE HEIGHT OF THE BALE.

EROSION CONTROL MATTING - REPAIR MATTING IMMEDIATELY IF INSPECTION REVEALS BREACHED OR FAILED CONDITIONS. REPAIR AND RE-GRADE SOIL WHERE CHANNELIZATION HAS OCCURRED.

DIVERSION BERM/SWALE - REPLACE OR RE-COMPACT THE CONSTRUCTION MATERIALS AS NECESSARY.

SEDIMENT TRAP - REMOVE AND DISPOSE OF THE ACCUMULATED SEDIMENT WHEN IT HAS REACHED THE SEDIMENT STORAGE ELEVATION.

INLET PROTECTION - CLEAN, REPAIR OR REPLACE FILTER FABRIC AND/OR STONE WHEN CONTROL MEASURE IS CLOGGED. INLET FILTER BAGS SHALL MBE REPLACED ONCE ONE-HALF FULL OF SEDIMENT.

OUTLET PROTECTION - CLEAN, REPAIR OR REPLACE FILTER FABRIC, TURF REINFORCEMENT MATTING AND/OR STONE WHEN CONTROL MEASURE IS ONE-HALF FULL OF SEDIMENT.

SEDIMENT BASIN - AT THE END OF CONSTRUCTION, CONTRACTOR SHALL REMOVE AND DISPOSE OF THE ACCUMULATED SEDIMENT AND RESTORE BASIN AREA TO INTENDED POST-CONSTRUCTION DESIGN GRADES.

Table 1 – Prescriptive Compliance Area Soil Stabilization

Prescriptive Compliance Areas	Bare Soil	Slope & Channel Management	Periods of Inactivity	Final Grade
Soil stockpiles that will exist for more than 7 days Utility trench backfills Temporary ditches/swales that will exist for more than 7 days Permanent ditches/swales Small areas – Less than 1 acre and less than 1% of site Discrete areas – Less than 1 acre Storm water practice side slopes Slopes steeper than 20%	Areas that Do Not Drain to Sediment Basins or Traps Limit the duration of soil exposure to no more than 30 days. Areas that Drain to Sediment Basins or Traps Limit the duration of soil exposure to no more than 90 days. However, use the duration from the soil loss and sediment discharge calculations for the other areas of the site if less than 90 days.	General Design and implement approved soil stabilization practices per DNR technical standards. Refer to WisDOT Slope & Channel Matrices for appropriate slope and slope length conditions. Slopes Steeper than 20% Provide stable diversion of off-site runoff around the slope. Provide slope interruption devices in accordance with Manufactured Perimeter Control & Slope Interruption Products Technical Standard 1071 or equivalent methods to reduce uninterrupted slope length.	Planned Inactivity Stabilize immediately if area will be left inactive for more than 14 days. Unplanned Inactivity Stabilize area immediately if period of inactivity reaches 14 days.	Permanent Features Stabilize area immediately after reaching final grade. Temporary Features Stabilize area immediately after establishment of temporary feature or reaching specified temporary grade.

5.0 INSPECTION

INSPECTIONS SHALL BE COMPLETED WITHIN TWENTY-FOUR (24) HOURS OF THE END OF A RAINFALL EVENT THAT IS ONE-HALF INCH OR GREATER OR EQUIVALENT SNOWFALL, OR AT A MINIMUM ONCE EVERY SEVEN (7) CALENDAR DAYS. INSPECTIONS SHALL BE UNDERTAKEN BY QUALIFIED PERSONNEL PROVIDED BY THE CONTRACTOR, AND SHALL INCLUDE: DISTURBED AREAS OF THE CONSTRUCTION SITE THAT HAVE NOT BEEN FINALLY STABILIZED, STRUCTURAL CONTROL MEASURES, AND LOCATIONS WHERE VEHICLES ENTER OR EXIT THE SITE. A STORMWATER POLLUTION PREVENTION PLAN INSPECTION REPORT SHALL BE COMPLETED AND ADDED TO THE SWPPP. RAINFALL SHALL BE RECORDED ON THE SWPPP RAINFALL LOG. CONTRACTOR SHALL IMMEDIATELY ARRANGE FOR REPAIR OR REPLACEMENT OF ANY DAMAGED OR DEFICIENT CONTROL MEASURES OBSERVED DURING THE INSPECTION.

QUALIFIED PERSONNEL MEANS A PERSON KNOWLEDGEABLE IN THE PRINCIPLES AND PRACTICES OF EROSION AND SEDIMENT CONTROL MEASURES, SUCH AS A LICENSED PROFESSIONAL ENGINEER, A CERTIFIED PROFESSIONAL IN EROSION AND SEDIMENT CONTROL, A CERTIFIED EROSION SEDIMENT OR STORMWATER INSPECTOR, OR OTHER TRAINED INDIVIDUAL.

6.0 SPILL PREVENTION

6.1 GENERAL MATERIAL MANAGEMENT PRACTICES

THE GOOD HOUSEKEEPING PRACTICES LISTED BELOW SHALL BE FOLLOWED THROUGHOUT THE CONSTRUCTION PROJECT.

1. CONTRACTOR SHALL STORE ONLY ENOUGH PRODUCTS REQUIRED TO COMPLETE THIS PROJECT.
2. ALL MATERIAL SHALL BE STORED IN A NEAT, ORDERLY MANNER IN THEIR ORIGINAL CONTAINERS CONTAINING MANUFACTURER'S LABEL.
3. MANUFACTURERS' RECOMMENDATIONS FOR PROPER USE AND DISPOSAL SHALL BE FOLLOWED.
4. MATERIALS REQUIRED TO HAVE A MATERIAL SAFETY DATA SHEET (MSDS) SHALL HAVE A COPY STORED IN THE PROJECT'S MSDS DATABASE.

6.2 SPILL CONTROL PRACTICES

THE PRACTICES LISTED BELOW SHALL BE FOLLOWED FOR SPILL PREVENTION AND CLEANUP.

1. MATERIALS AND EQUIPMENT NECESSARY FOR SPILL CLEANUP SHALL BE MAINTAINED ONSITE.
2. IMMEDIATELY UPON DISCOVERY, ALL SPILLS SHALL BE CLEANED UP ACCORDING TO THE MANUFACTURERS' RECOMMENDED METHODS.
3. PERSONNEL CLEANING UP A SPILL SHALL USE PERSONAL PROTECTIVE EQUIPMENT.
4. IMMEDIATELY UPON DISCOVERY, SPILLS OF TOXIC OR HAZARDOUS MATERIALS SHALL BE REPORTED TO THE OWNER AND GENERAL CONTRACTOR.
5. NOTIFICATION AND REPORTING TO THE APPROPRIATE FEDERAL, STATE, AND LOCAL GOVERNMENT AGENCIES SHALL BE MADE AS REQUIRED.

GENERAL INFORMATION:

THIS STORMWATER POLLUTION PREVENTION PLAN (SWPPP) HAS BEEN DEVELOPED TO FULFILL ONE OF THE REQUIREMENTS OF THE GENERAL ENVIRONMENTAL PROTECTION AGENCY (EPA) NATIONAL POLLUTION DISCHARGE ELIMINATION SYSTEM (WISCONSIN POLLUTION DISCHARGE ELIMINATION SYSTEM "WPDES" PERMIT NO. WI-S067831-4) FOR THE DISCHARGE OF STORMWATER ASSOCIATED WITH CONSTRUCTION PROJECTS DISTURBING ONE ACRE OR MORE. THE OWNER AND CONTRACTORS SHALL COMPLY WITH ALL REQUIREMENTS OF THE WPDES FOR ALL SUCH CONSTRUCTION PROJECTS. THE STORMWATER DISCHARGES ASSOCIATED WITH THE CONSTRUCTION ACTIVITY FROM THIS SITE ARE SUBJECT TO THE CONDITIONS AND REQUIREMENTS OF THE PERMITS.

THE EXECUTED OWNER CERTIFICATION AND THE CONTRACTOR CERTIFICATIONS SHALL BE KEPT ONSITE WITH THE APPROVED PLANS.

SWPPP AVAILABILITY:

THE OWNER SHALL RETAIN A COPY OF THE SWPPP AT THE CONSTRUCTION SITE FROM THE DATE OF THE PROJECT INITIATION TO THE DATE OF FINAL STABILIZATION.

KEEPING PLANS CURRENT:

THE CONTRACTOR SHALL AMEND THE PLAN WHENEVER THERE IS A CHANGE IN DESIGN, CONSTRUCTION, OPERATION, OR MAINTENANCE, WHICH HAS A SIGNIFICANT EFFECT ON THE POTENTIAL FOR THE DISCHARGE OF POLLUTANTS TO THE WATERS OF THE STATE AND WHICH HAS NOT OTHERWISE BEEN ADDRESSED IN THE PLAN OR IF THE PLAN PROVES TO BE INEFFECTIVE IN ELIMINATING OR SIGNIFICANTLY CONTROLLING POLLUTANTS IN STORMWATER DISCHARGES ASSOCIATED WITH CONSTRUCTION SITE ACTIVITY. IN ADDITION, THE THE PLAN SHALL BE AMENDED TO IDENTIFY ANY NEW CONTRACTOR AND/OR SUBCONTRACTOR THAT WILL IMPLEMENT A MEASURE OF THE PLAN. AMENDMENTS TO THE PLAN MAY BE REQUIRED BY THE MUNICIPALITY, OWNER, OR OTHER REVIEWING AGENCY. COPIES OF THE AMENDMENTS SHALL BE KEPT ONSITE AS PART OF THE SWPPP.

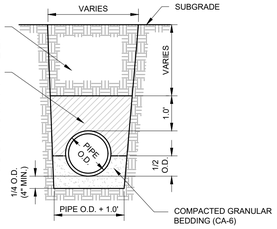
RETENTION OF RECORDS:

THE OWNER SHALL RETAIN COPIES OF THIS AND ALL REPORTS AND NOTICES REQUIRED BY THIS PERMIT, AND RECORDS OF ALL DATA USED TO COMPLETE THE NOTICE OF INTENT TO BE COVERED BY THIS PERMIT, FOR A PERIOD OF AT LEAST THREE YEARS FROM THE DATE PERMIT COVERAGE EXPIRES OR IS TERMINATED. THIS PERIOD MAY BE EXTENDED BY THE REQUEST OF THE AGENCY AT ANY TIME. IN ADDITION, THE OWNER SHALL RETAIN A COPY OF THE PLAN REQUIRED BY THIS PERMIT AT THE CONSTRUCTION SITE FROM THE DATE OF PROJECT INITIATION TO THE DATE OF FINAL STABILIZATION.

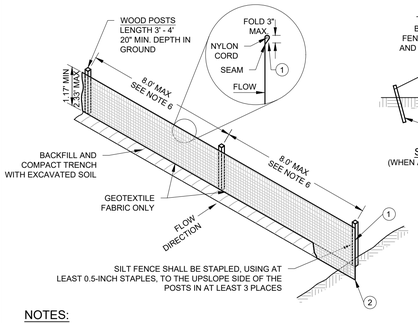
A NOTICE OF INTENT (NOI) APPLICATION MUST BE COMPLETED AND INCORPORATED INTO THE SWPPP.

WPDES NOTICE OF TERMINATION GUIDANCE:

WHEN A SITE HAS BEEN FINALLY STABILIZED AND ALL STORMWATER DISCHARGES FROM CONSTRUCTION SITES THAT ARE AUTHORIZED BY THE PERMIT ARE ELIMINATED, THE OWNER OF THE FACILITY MUST SUBMIT A COMPLETED NOTICE OF TERMINATION THAT IS SIGNED IN ACCORDANCE WITH THE PERMIT. CONTRACTOR SHALL SUBMIT A COMPLETED NOTICE OF TERMINATION TO OWNER FOR EXECUTION PRIOR TO THEIR FINAL PAY APPLICATION REQUEST.

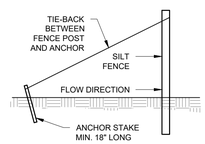


SEWER AND WATERMAIN BEDDING & BACKFILL

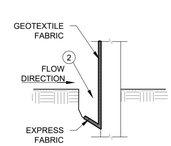


NOTES:

1. GEOTEXTILE FABRIC SHALL MEET THE REQUIREMENTS OF MATERIAL SPECIFICATION 592 GEOTEXTILE TABLE 1 OR 2, CLASS I WITH EQUIVALENT OPENING SIZE OF AT LEAST 30 FOR NONWOVEN AND 50 FOR WOVEN.
2. SILT FENCE SHALL BE ANCHORED BY SPREADING AT LEAST 8-INCHES OF FABRIC IN A 4-INCH WIDE AND 6-INCH DEEP TRENCH OR 6-INCH DEEP V-TRENCH ON THE UPSLOPE SIDE OF THE FENCE. TRENCHES SHALL NOT BE EXCAVATED WIDER OR DEEPER THAN NECESSARY FOR PROPER INSTALLATION.
3. FOLD MATERIAL TO FIT TRENCH AND BACKFILL AND COMPACT TRENCH WITH EXCAVATED SOIL.
4. WOOD POSTS SHALL BE A MINIMUM SIZE OF 1.125-INCHES x 1.125-INCHES OF DRIED OAK OR HICKORY.
5. SILT FENCE TO EXTEND ABOVE THE TOP OF PIPE, WHERE APPLICABLE.
6. POST SPACING SHALL BE SELECTED BASED ON GEOTEXTILE FABRIC (8-FEET FOR WOVEN AND 3-FEET FOR NON-WOVEN).

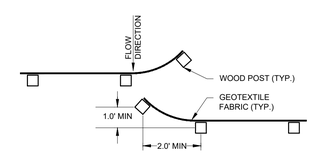


SILT FENCE TIE BACK
(WHEN ADDITIONAL SUPPORT REQUIRED)

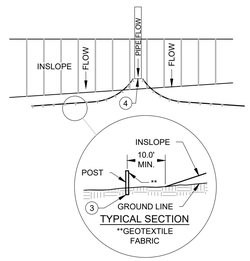


NOTE: ADDITIONAL POST DEPTH OR TIE BACKS MAY BE REQUIRED IN UNSTABLE SOILS

TRENCH DETAIL

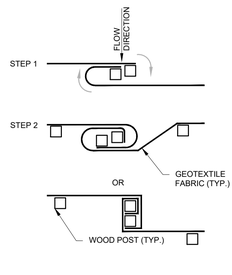


JOINT (HOOK METHOD)



TYPICAL SECTION
**GEOTEXTILE FABRIC

SILT FENCE ALONG SLOPES & OUTFALLS



JOINT (TWIST METHOD)

SILT FENCE

PLAN | DESIGN | DELIVER
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(262) 754-8888
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ZILBER INDUSTRIAL 3 @ LAKEVIEW CORPORATE PARK

108TH STREET & GREEN BAY ROAD (STH "31")
VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

CONSTRUCTION DETAILS

REVISIONS

1	VOPP SUBMITTAL	05/21/26	
2	VOPP RESUBMITTAL	07/17/26	
3	VOPP RESUBMITTAL	10/02/26	
4	BID SET/VOPP RESUBMITTAL	11/03/26	

PEG JOB No. 644.00-WI
PEG PM AJS
START DATE 03/20/26
SCALE N.T.S.

SHEET
C-6
C-8

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CONSTRUCTION DETAILS

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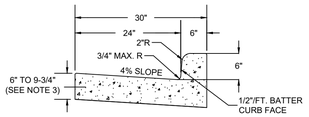
REVIEWED:
DESIGNED: AJS
DRAFTED: AJS

Z:\PROJECTS\2016\644.00-WI\CAD\SHEETS\WATERMAIN PLAN SHEETS\644.00-WI-SHEETS-CONSTRUCTION DETAILS.DWG

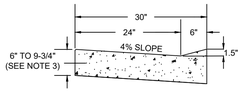
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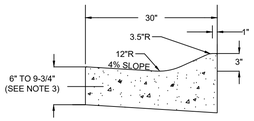
NOTES:
1. DAMAGED CURB / GUTTER SECTIONS SHALL BE REMOVED TO THE NEAREST JOINT.
2. CURB AND GUTTER CONSTRUCTED ADJACENT TO EXISTING CURB AND GUTTER SHALL BE INSTALLED USING TWO (2) NO.4 (1/2-INCH), 18-INCH LONG TIE BARS, EVENLY SPACED, DRIVEN 9-INCHES INTO THE EXISTING CURB AND GUTTER.
3. WHERE ADJACENT PAVEMENT SECTION CONTAINS CONCRETE THE GUTTER THICKNESS SHALL EXTEND TO THE BOTTOM OF THE ADJACENT CONCRETE PAVEMENT. CURB AND GUTTER SHALL BE TIED INTO ADJACENT COMPOSITE PAVEMENT WITH #4 X 2'0" LONG TIE BARS AT 3' SPACING.



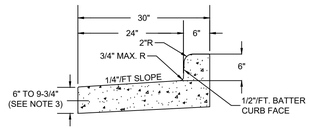
30" VERTICAL FACE CURB AND GUTTER



30" DEPRESSED CURB AND GUTTER AT DRIVEWAYS



30" MOUNTABLE CURB AND GUTTER



30" VERTICAL FACE CURB AND GUTTER (REVERSE SLOPE GUTTER)

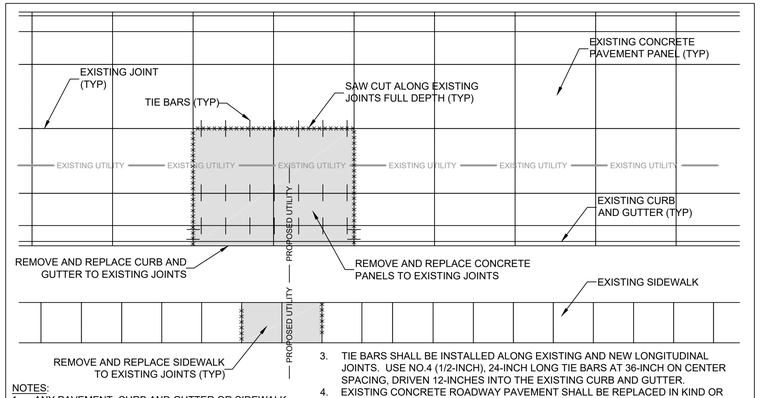
SCALE: NTS

STANDARD CURB & GUTTER DETAILS

DETAIL: RD - 9

DATE: 6-5-20

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



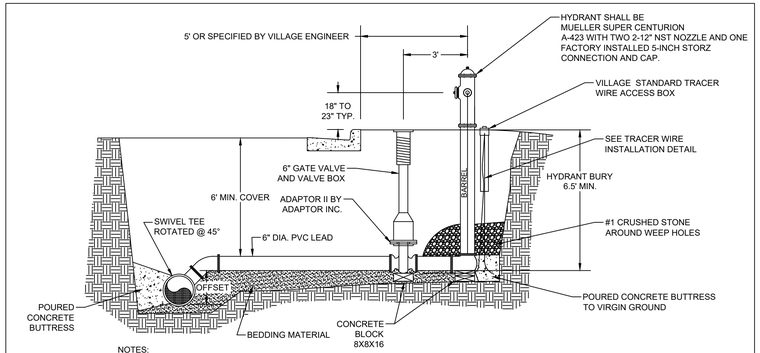
NOTES:
1. ANY PAVEMENT, CURB AND GUTTER OR SIDEWALK DAMAGED OR UNDERMINED SHALL BE REMOVED AND REPLACED TO THE NEAREST JOINT.
2. CURB AND GUTTER SHALL BE INSTALLED USING TWO (2) NO.4 (1/2-INCH), 18-INCH LONG TIE BARS, EVENLY SPACED, DRIVEN 9-INCHES INTO THE EXISTING CURB AND GUTTER.
3. TIE BARS SHALL BE INSTALLED ALONG EXISTING AND NEW LONGITUDINAL JOINTS. USE NO.4 (1/2-INCH), 24-INCH LONG TIE BARS AT 36-INCH ON CENTER SPACING, DRIVEN 12-INCHES INTO THE EXISTING CURB AND GUTTER.
4. EXISTING CONCRETE ROADWAY PAVEMENT SHALL BE REPLACED IN KIND OR WITH A MINIMUM OF 8-INCHES OF BASE AGGREGATE DENSE AND 8-INCHES OF CONCRETE PAVEMENT, WHICHEVER IS GREATER.
5. EXISTING COMPOSITE ROADWAY PAVEMENT SHALL BE REPLACED IN KIND OR WITH A MINIMUM OF 8-INCHES OF BASE AGGREGATE DENSE, 7-INCHES OF CONCRETE PAVEMENT (8-INCHES ON INDUSTRIAL ROAD AND COLLECTORS), AND 1.75-INCHES OF ASPHALTIC CONCRETE SURFACE, WHICHEVER IS GREATER.
6. REFER TO VILLAGE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

UTILITY PATCH DETAIL (COMPOSITE AND CONCRETE ROADWAYS)

DETAIL: RD - 15

DATE: 12-17-18

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



MAIN DIA (IN.)	OFFSET (FT.)
16	1.6±
18	1.7±
20	1.9±
24	2.2±

NOTES:
1. MECHANICAL JOINTS FROM TEE TO VALVE AND FROM VALVE TO HYDRANT SHALL BE RESTRAINED WITH MEGALUGS AND STAINLESS STEEL BOLTS.
2. ALL FASTENERS SHALL BE STAINLESS STEEL.
3. HYDRANT EXTENSIONS ARE NOT PERMITTED.
4. ALL HYDRANTS, EXCLUDING STORZ CONNECTION, SHALL BE FACTOR PAINTED WITH EPOXY AS IDENTIFIED BELOW:
   A. PUBLIC (VILLAGE) HYDRANTS SHALL BE MUELLER RED - SHERWIN WILLIAMS POLANE F63RL15.
   B. PRIVATE HYDRANTS ON VILLAGE SYSTEM PRESSURE SHALL BE CAUTION BLUE - SHERWIN WILLIAMS POLANE F63RXL9667.
   C. PRIVATE PRESSURIZED HYDRANTS ON FIRE LOOPS SHALL BE MUELLER YELLOW - SHERWIN WILLIAMS POLANE F63YL14.
5. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN CONSTRUCTION.

SCALE: NTS

AIR RELEASE HYDRANT ASSEMBLY

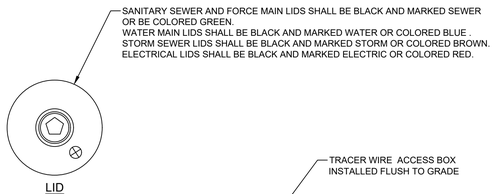
DETAIL: W - 2

DATE: 8-22-19

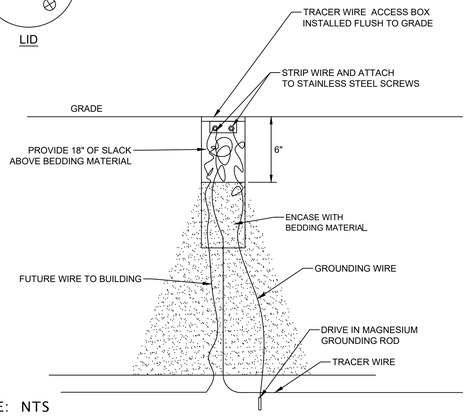
VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

NOTES

1. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR TRACER WIRE.



LID



SCALE: NTS

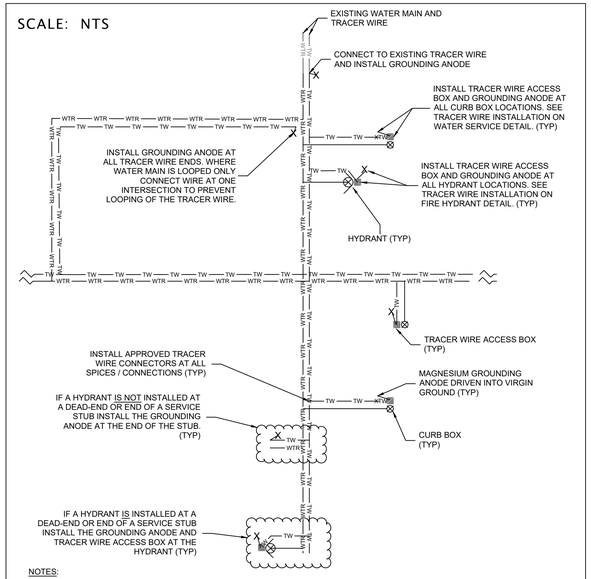
TRACER WIRE ACCESS BOX

DETAIL: TW - 1

DATE: 2-4-2025

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

SCALE: NTS



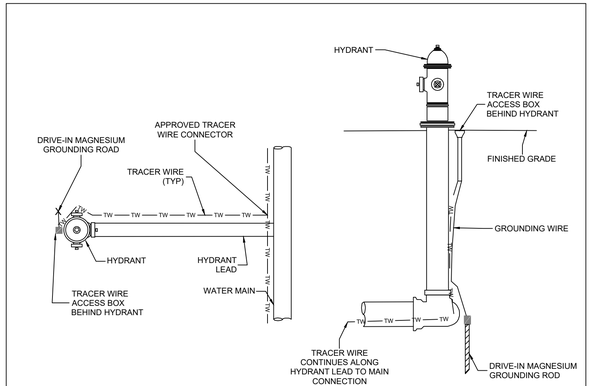
NOTES:
1. REFER TO SPECIFICATIONS FOR TRACER WIRE.
2. CONNECT TO EXISTING TRACER WIRE AT ONLY ONE LOCATION UNLESS OTHERWISE DIRECTED BY THE VILLAGE.
3. TRACER WIRE SHALL BE TAPED TO MAIN AND SERVICES. WIRE IS SHOWN OFF PIPES FOR CLARITY.
4. TAKE CARE TO PREVENT LOOPS IN TRACER WIRE. LOOPING OF WIRE CAN MAKE THE WIRE DIFFICULT TO TRACE.

TRACER WIRE INSTALLATION ON WATER MAIN

DETAIL: TW - 3

DATE: 12-17-18

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



PLAN VIEW SECTION VIEW

NOTES:
1. REFER TO SPECIFICATIONS FOR TRACER WIRE.
2. TRACER WIRE SHALL BE TAPED TO MAIN, LEAD AND HYDRANT. WIRE IS SHOWN OFF PIPES FOR CLARITY.
3. INSTALL GROUNDING ANODE BY DRIVING INTO VIRGIN GROUND AT THE HYDRANT SHOE.
4. TRACER WIRE ACCESS BOX SHALL BE INSTALLED DIRECTLY BEHIND ALL FIRE HYDRANTS.

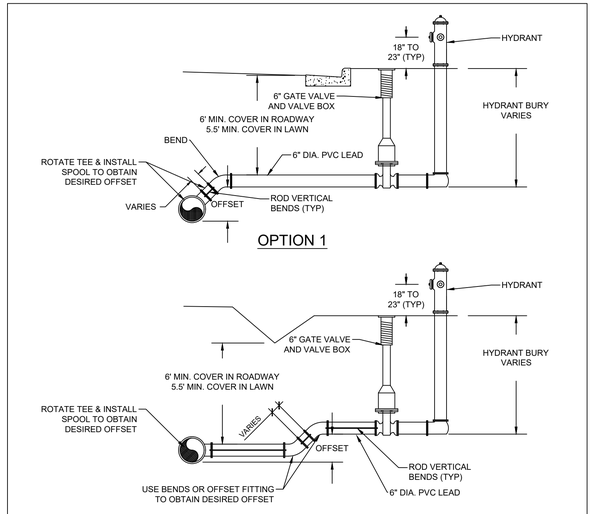
SCALE: NTS

TRACER WIRE INSTALLATION ON FIRE HYDRANT

DETAIL: TW - 4

DATE: 12-17-18

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



OPTION 2

NOTES:
1. HYDRANT ASSEMBLY INSTALLATION SHALL BE IN ACCORDANCE WITH THE STANDARD HYDRANT ASSEMBLY DETAIL.
2. REFER TO THE VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN CONSTRUCTION.
3. ALL VERTICAL BENDS SHALL BE RODDED WITH STAINLESS STEEL HARDWARE.

SCALE: NTS

HYDRANT OFFSETS

DETAIL: W - 6

DATE: 8-4-16

APPROVED BY: MATT FINEOUR

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

REVIEWED:
DESIGNED: AJS
DRAFTED: AJS

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20725 WATERTOWN ROAD, SUITE 100
BROOKFIELD, WI 53186
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108TH STREET & GREEN BAY ROAD (STH "31")
VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

CONSTRUCTION DETAILS

REVISIONS

1	VOPP SUBMITTAL	05/21/26		
2	VOPP RESUBMITTAL	07/17/26		
3	VOPP RESUBMITTAL	10/02/26		
4	BID SET/VOPP RESUBMITTAL	11/03/26		

PEG JOB No. 644.00-WI
PEG PM AJS
START DATE 03/20/26
SCALE N.T.S.

SHEET
C-7
C-8

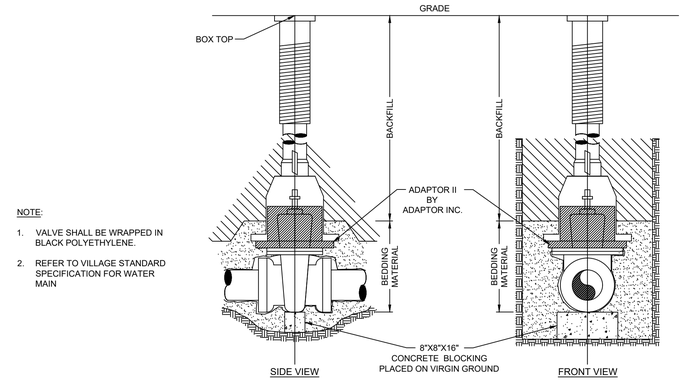
CONSTRUCTION DETAILS

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NOTE:

1. VALVE SHALL BE WRAPPED IN BLACK POLYETHYLENE.
2. REFER TO VILLAGE STANDARD SPECIFICATION FOR WATER MAIN

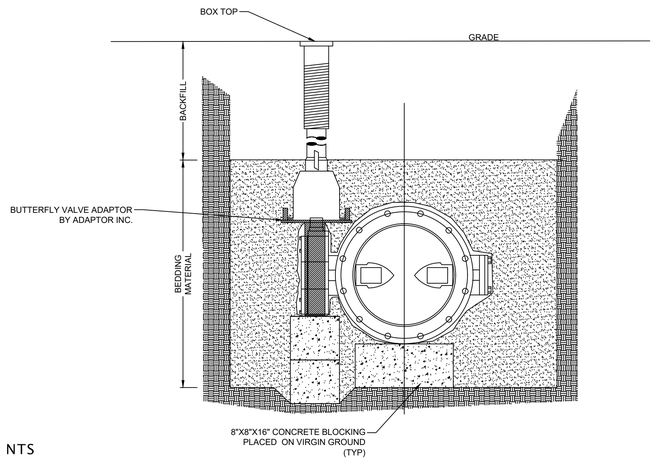
SCALE: NTS

STANDARD GATE VALVE BOX SETTING

DETAIL: W - 3

DATE: 11-16-15

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



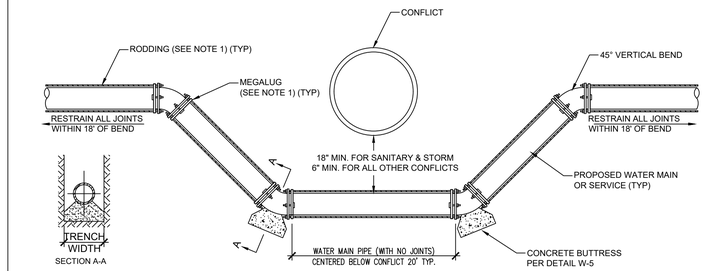
SCALE: NTS

STANDARD BUTTERFLY VALVE BOX SETTING

DETAIL: W - 4

DATE: 11-16-15

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN



NOTES:

1. VILLAGE OF PLEASANT PRAIRIE PUBLIC WATER MAINS - ALL JOINTS SHALL BE RESTRAINED WITH MEGALUGS OR STAINLESS STEEL RODDING IN CONFORMANCE WITH FILE NO. 47 OF THE "STANDARD SPECIFICATIONS." PRIVATE FIRE MAINS AND COMBINATION FIRE/WATER MAINS - ALL JOINTS SHALL BE RESTRAINED WITH MEGALUGS AND STAINLESS STEEL RODDING IN CONFORMANCE WITH SUBSECTION 10.6.2.1.2 OF THE NFPA 24 STANDARD FOR THE INSTALLATION OF PRIVATE SERVICE MAINS AND THEIR APPURTENANCES.
2. ALL FASTENERS SHALL BE STAINLESS STEEL .
3. WHEN NOTED ON THE PLANS OR DIRECTED BY THE VILLAGE, INSULATION SHALL BE INSTALLED BETWEEN THE WATER MAIN AND CONFLICT.
4. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN CONSTRUCTION

SCALE: NTS

WATER MAIN VERTICAL OFFSET

DETAIL: W - 9

DATE: 5-24-21

APPROVED BY: MATT FINEOUR

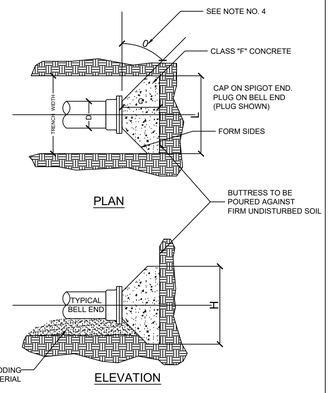
VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

NOTES:

1. $L_{(WATER)}$ DIMENSIONS IN THE TABLE ARE BASED ON A WATER PRESSURE OF 150 PSI, AN EARTH RESISTANCE OF 2 TONS PER SQ. FT., A FACTOR OF SAFETY OF 1.5 AND IS TO BE USED FOR STANDARD WATER MAINS AND WATER SERVICES.
2. $L_{(FIRE)}$ DIMENSIONS IN THE TABLE ARE BASED ON A WATER PRESSURE OF 200 PSI, AN EARTH RESISTANCE OF 2 TONS PER SQ. FT., A FACTOR OF SAFETY OF 1.5 AND IS TO BE USED FOR FIRE LINES AND COMBINATION FIRE / WATER SERVICES.
3. SHAPE OF BACK OF BUTTRESS MAY VARY AS LONG AS POUR IS AGAINST FIRM UNDISTURBED EARTH AND SURFACE AREA (H x L) MATCHES THAT LISTED IN THE TABLE.
4. DIMENSION "C" SHOULD BE LARGE ENOUGH TO MAKE ANGLE "0" EQUAL TO OR LARGER THAN 45 DEG.
5. DIMENSION "D" EQUALS APPROX. I.D. OF PIPE LESS 2". AN EFFORT SHOULD BE MADE TO PREVENT CONCRETE FROM COVERING THE MECHANICAL JOINT BOLTS.
6. WHERE BUTTRESSES ARE NOT POSSIBLE BECAUSE OF POOR SOIL CONDITIONS OR LACK OF ROOM, STRAPPING SHALL BE PERMITTED IF APPROVED BY THE VILLAGE.
7. IF IN-SITU SOILS HAVE AN EARTH RESISTANCE OF LESS THAN 2 TONS PER SQ. FT. THE BUTTRESS AREA SHALL BE INCREASED PROPORTIONALLY.
8. ALL DUCTILE IRON PIPE AND FITTINGS SHALL BE WRAPPED IN POLYETHYLENE.
9. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN.

BUTTRESS DIMENSIONS

PIPE	DEAD ENDS		
SIZE	H	$L_{(WATER)}$	$L_{(FIRE)}$
4"	1'-6"	1'-0"	1'-0"
6"	1'-8"	1'-9"	1'-9"
8"	2'-4"	2'-0"	2'-1"
10"	2'-4"	2'-4"	3'-1"
12"	2'-7"	3'-0"	4'-0"
14"	2'-10"	3'-8"	4'-10"
16"	2'-10"	4'-9"	6'-4"
18"	3'-6"	4'-10"	6'-5"
20"	3'-10"	5'-5"	7'-2"
24"	4'-8"	6'-2"	8'-3"



SCALE: NTS

BUTTRESS FOR DEAD ENDS

DETAIL: W - 5B

DATE: 8-4-16

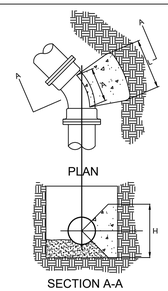
APPROVED BY: MATT FINEOUR

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

SCALE: NTS

NOTES:

1. $L_{(WATER)}$ DIMENSIONS IN THE TABLE ARE BASED ON A WATER PRESSURE OF 150 PSI, AN EARTH RESISTANCE OF 2 TONS PER SQ. FT., A FACTOR OF SAFETY OF 1.5 AND IS TO BE USED FOR STANDARD WATER MAINS AND WATER SERVICES.
2. $L_{(FIRE)}$ DIMENSIONS IN THE TABLE ARE BASED ON A WATER PRESSURE OF 200 PSI, AN EARTH RESISTANCE OF 2 TONS PER SQ. FT., A FACTOR OF SAFETY OF 1.5 AND IS TO BE USED FOR FIRE LINES AND COMBINATION FIRE / WATER SERVICES.
3. DIMENSION (A) SHOULD BE AS LARGE AS POSSIBLE WITHOUT INTERFERING WITH MECHANICAL JOINT BOLTS.
4. SHAPE OF BACK OF BUTTRESS MAY VARY AS LONG AS POUR IS AGAINST FIRM UNDISTURBED EARTH AND SURFACE AREA (H x L) MATCHES THAT LISTED IN THE TABLE.
5. IF IN-SITU SOILS HAVE AN EARTH RESISTANCE OF LESS THAN 2 TONS PER SQ. FT. THE BUTTRESS AREA SHALL BE INCREASED PROPORTIONALLY.
6. ALL DUCTILE IRON PIPE AND FITTINGS SHALL BE WRAPPED IN POLYETHYLENE.
7. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN.



BUTTRESS DIMENSIONS

PIPE	11¼ DEG. BEND			22½ DEG. BEND			45 DEG. BEND			90 DEG. BEND		
SIZE	H	$L_{(WATER)}$	$L_{(FIRE)}$	H	$L_{(WATER)}$	$L_{(FIRE)}$	H	$L_{(WATER)}$	$L_{(FIRE)}$	H	$L_{(WATER)}$	$L_{(FIRE)}$
4"	0'-8"	0'-5"	0'-5"	1'-1"	0'-6"	0'-6"	1'-3"	0'-10"	0'-10"	1'-4"	1'-6"	1'-6"
6"	1'-2"	0'-6"	0'-6"	1'-6"	0'-9"	0'-9"	1'-8"	1'-4"	1'-4"	1'-10"	2'-2"	2'-2"
8"	1'-3"	0'-9"	0'-9"	2'-0"	1'-0"	1'-0"	2'-0"	1'-11"	1'-11"	2'-4"	3'-0"	3'-0"
10"	1'-3"	0'-10"	1'-2"	2'-0"	1'-1"	1'-5"	2'-0"	2'-1"	2'-9"	2'-4"	3'-4"	4'-5"
12"	1'-4"	1'-2"	1'-6"	2'-6"	1'-2"	1'-7"	2'-6"	2'-4"	3'-2"	2'-8"	4'-1"	5'-5"
14"	1'-6"	1'-4"	1'-10"	3'-0"	1'-4"	1'-10"	3'-0"	2'-8"	3'-6"	3'-0"	4'-11"	6'-6"
16"	2'-0"	1'-4"	1'-9"	3'-0"	1'-9"	2'-4"	3'-0"	3'-5"	4'-7"	3'-0"	6'-4"	8'-5"
18"	2'-6"	1'-4"	1'-9"	3'-6"	1'-10"	2'-6"	3'-6"	3'-8"	4'-11"	3'-6"	6'-9"	9'-1"
20"	2'-9"	1'-6"	2'-0"	3'-9"	2'-2"	2'-10"	3'-9"	4'-2"	5'-7"	3'-9"	7'-9"	10'-4"
24"	3'-0"	1'-11"	2'-7"	4'-3"	2'-8"	3'-7"	4'-3"	5'-4"	7'-1"	4'-4"	9'-7"	12'-10"

BUTTRESS FOR BENDS

DETAIL: W - 5

DATE: 8-4-16

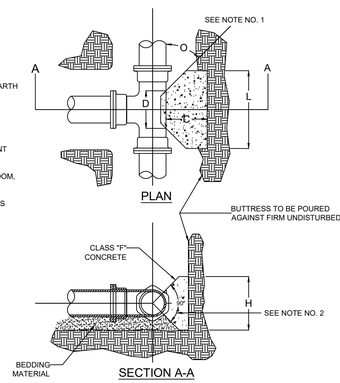
APPROVED BY: MATT FINEOUR

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

NOTES:

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3. SHAPE OF BACK OF BUTTRESS MAY VARY AS LONG AS POUR IS AGAINST FIRM UNDISTURBED EARTH AND SURFACE AREA (H x L) MATCHES THAT LISTED IN THE TABLE. CONCRETE SHALL BEAR ON FITTINGS AS SHOWN.
4. DIMENSION "C" SHOULD BE LARGE ENOUGH TO MAKE ANGLE "0" EQUAL TO OR LARGER THAN 45 DEG.
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8. ALL DUCTILE IRON PIPE AND FITTINGS SHALL BE WRAPPED IN POLYETHYLENE.
9. REFER TO VILLAGE STANDARD SPECIFICATIONS FOR WATER MAIN.

PIPE	TEE		
SIZE	H	$L_{(WATER)}$	$L_{(FIRE)}$
4"	1'-6"	1'-0"	1'-0"
6"	1'-8"	1'-9"	1'-9"
8"	2'-4"	2'-0"	2'-1"
10"	2'-4"	2'-4"	3'-1"
12"	2'-7"	3'-0"	4'-0"
14"	2'-10"	3'-8"	4'-10"
16"	2'-10"	4'-9"	6'-4"
18"	3'-6"	4'-10"	6'-5"
20"	3'-10"	5'-5"	7'-2"
24"	4'-8"	6'-2"	8'-3"



SCALE: NTS

BUTTRESS FOR TEES

DETAIL: W - 5A

DATE: 8-4-16

VILLAGE OF PLEASANT PRAIRIE
KENOSHA COUNTY, WISCONSIN

REVISED:

DESIGNED: AJS

DRAFTED: AJS

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VILLAGE OF PLEASANT PRAIRIE, KENOSHA COUNTY

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1	VOPP SUBMITTAL	05/21/26			
2	VOPP RESUBMITTAL	07/17/26			
3	VOPP RESUBMITTAL	10/02/26			
4	BID SET/VOPP RESUBMITTAL	11/03/26			

PEG JOB No. 644.00-WI
PES PM AJS
START DATE 03/20/26
SCALE N.T.S.

SHEET
C-8
C-8

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CONSTRUCTION DETAILS

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